Privacy, Utility, and Explainability Trade-offs in Privacy-Preserving Federated Recommender Systems

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Abstract

Purpose: This study provides a controlled empirical characterization of privacy, utility, system efficiency, and explainability trade-offs in federated recommender systems, evaluating how Differential Privacy (DP) and Homomorphic Encryption (HE) affect ranking accuracy, infrastructure costs, and post-hoc explanations.

Design/methodology/approach: We evaluate centralized training, plaintext Federated Averaging (FedAvg), three client-level DP noise settings ($\sigma \in \{0.5, 1.0, 2.0\}$), TenSEAL CKKS homomorphic aggregation, and hybrid DP+HE within the multi-branch `CoreHybridRecSys` architecture. Training exposure is aligned at 19,206 interaction samples (3.724 passes over 5,157 partitioned EARS training samples across 20 clients). We track five ranking metrics, training and aggregation runtimes, communication payloads, dual XAI attribution measures (KernelSHAP vs LIME), and empirical privacy attack resilience (DLG gradient inversion and MIA membership inference) across ten paired random seeds $\{42, \ldots, 51\}$, complemented by second-dataset validation on MovieLens-100k.

Findings: CKKS homomorphic aggregation preserves ranking utility with practical equivalence to plaintext FedAvg (EARS NDCG@10 of 0.5377 ± 0.0279 vs 0.5379 ± 0.0268, paired difference -0.00022 ± 0.00289 within a ±0.01 margin; Two One-Sided Tests $p_{\text{TOST}} = 1.01 \times 10^{-6}$ with 100% power, and $p_{\text{TOST}} = 2.68 \times 10^{-4}$ under ±0.005 with $df = 9$), but incurs 24.52 ± 0.29 s server aggregation latency and a 21.51× client communication expansion (3,610.90 MB). We demonstrate that Top-K parameter sparsification ($K = 10\%$) eliminates 90.0% of communication payload while preserving ranking accuracy. Empirical attack validation reveals that plaintext FedAvg suffers 100% gradient inversion reconstruction, whereas CKKS reduces reconstruction success to 0%, while DP bounds membership inference leakage. Differential privacy induces severe ranking utility degradation (0.8474 → 0.22–0.35 Precision@10) and disrupts post-hoc explanation fidelity. Under the configured evaluation budgets (1,500 model calls for KernelSHAP vs 15,000 for LIME), KernelSHAP exhibited 5 × –7× lower per-instance latency (0.52–0.72 s vs 3.41–3.76 s). Client-blocked stratified batching reduced centralized runtime from 801.34 s to 27.32 ± 1.83 s, demonstrating that data-access locality, rather than algorithmic FL superiority, governs training throughput.

Originality: Unlike prior studies evaluating privacy, cryptography, or explainability in isolation, this work provides a joint empirical characterization across all four dimensions, delivering an actionable multi-criteria decision framework and cloud TCO cost model for organizational data sovereignty and digital information governance.

Keywords: Federated Learning; Privacy-Preserving Machine Learning; Differential Privacy; Homomorphic Encryption; Explainable AI; Recommender Systems.
Paper type: Research paper.

1 Introduction

1.1 Background and Motivation

Recommender systems support product discovery and user engagement across e-commerce, streaming media, and digital services (Koren et al., 2009; He et al., 2017; Covington et al., 2016). Many platforms rely on centralized aggregation of customer profiles, browsing histories, click streams, and transaction logs. Although this architecture simplifies model optimization, it also concentrates sensitive behavioral

data and may conflict with privacy, data-governance, and organizational sovereignty requirements (Yang et al., 2019; Gaspar, Cruz, Rodrigues, Rodrigues, Ribeiro, Martinho, Santos and Marreiros, 2025).

Federated Learning (FL) (McMahan et al., 2017; Kairouz et al., 2021) provides an alternative in which model training is distributed across independent nodes, such as regional enterprise branches, partner retailers, or edge devices. In Horizontal Federated Learning (HFL), clients share a common feature space but maintain disjoint user interaction sets. Each participant trains a local model replica and sends parameter updates, rather than raw records, to a coordinating server. The server commonly combines these updates through Federated Averaging (FedAvg) (McMahan et al., 2017).

The baseline relies on the open-source Environment Adaptive Recommendation System (EARS) architecture (Gaspar, Cruz, Rodrigues, Rodrigues, Ribeiro, Martinho, Santos and Marreiros, 2025; Gaspar et al., 2026), where FL supports collaborative model training across organizational data holders without pooling interaction records at a single coordinating server.

1.2 The Privacy–Utility–Explainability Trilemma

Keeping raw records local reduces direct data transfer, but unprotected model updates can still reveal information about client data. Prior theoretical literature identifies several prominent threat vectors:

1. **Gradient Inversion Attacks:** An untrusted or honest-but-curious server may attempt to reconstruct private training instances by solving an optimization problem matching synthesized gradients to observed client updates (Javeed et al., 2023; Gaspar, Pessoa, Ribeiro, Martinho, Santos and Marreiros, 2025).

2. **Membership and Attribute Inference:** Adversaries may infer whether specific individuals or cohorts participated in a client's dataset or extract private user attributes by observing global model updates (Abadi et al., 2016).

In this study, privacy-preserving mechanisms are incorporated to address these theoretical threat models and are empirically validated against canonical attack formulations. Specifically, we evaluate both theoretical privacy guarantees and empirical attack resilience (Deep Leakage from Gradients and Membership Inference Attacks; Table V):

- **Differential Privacy (DP):** Bounds the influence of a declared privacy unit through update clipping and calibrated Gaussian noise injection. The unit depends on the training design; the client-update mechanism evaluated here is designed to protect a client partition, not an individual interaction or record (Dwork et al., 2006; Abadi et al., 2016).

- **Homomorphic Encryption (HE):** Enables mathematical operations (specifically additions and scalar multiplications in FedAvg) to be performed directly on encrypted ciphertexts using schemes such as Cheon-Kim-Kim-Song (CKKS) (Cheon et al., 2017; Benaissa et al., 2021), preventing the server from inspecting individual updates in plaintext.

Recommender systems must also support auditability, transparency, and intelligible decision analysis, particularly when their outputs influence consequential organizational or commercial processes (Zhang and Chen, 2020). In this study, post-hoc feature attribution is implemented with SHAP (SHapley Additive exPlanations) (Lundberg and Lee, 2017) and LIME (Local Interpretable Model-agnostic Explanations) (Ribeiro et al., 2016).

These requirements create complex, multi-dimensional trade-offs among privacy, ranking utility, computational cost, communication volume, and explanation behavior. Differential privacy introduces stochastic perturbation that degrades ranking utility and alters feature attribution, whereas homomorphic encryption preserves arithmetic precision at the expense of substantial cryptographic computation and ciphertext communication expansion. Prior studies have typically evaluated these dimensions in isolation—such as studying DP utility loss without encryption overhead, or assessing HE latency without explainability impact.

1.3 Hypotheses and Research Questions

To guide the evaluation, we formulate four hypotheses that are assessed using the reported experimental measurements:

- **H_1** (*CKKS Ranking Practical Equivalence*): Under the configured CKKS precision (2^{40} scale), ranking measurements for encrypted aggregation will fall within a practical equivalence margin of $\Delta_{\text{equiv}} = \pm 0.01$ in NDCG@10 relative to plaintext FedAvg.
- **H_2** (*Ranking Utility Under Differential Privacy*): Differential privacy noise injection will cause substantial ranking utility degradation relative to plaintext FedAvg, scaling with the nominal noise scale $\sigma \in \{0.5, 1.0, 2.0\}$ across macroscopic sweep boundaries.
- **H_3** (*XAI Operational Latency under Configured Budgets*): Under the specified, method-specific explainer budgets (1,500 model evaluations for KernelSHAP versus 15,000 for LIME across 30 instances), KernelSHAP will exhibit lower measured runtime per explanation than LIME.
- **H_4** (*Attribution Alignment under Differential Privacy*): DP configurations will show lower attribution alignment with the centralized reference (E0) than plaintext federated learning, as measured by attribution cosine similarity and Top-5 feature agreement.

These hypotheses lead to four formal research questions:

- **RQ1:** How do different privacy configurations (FedAvg, multi-tier DP, CKKS HE, and hybrid DP+HE) affect top-K recommendation ranking quality (Precision, Recall, NDCG, MAP, Hit Rate) compared with a centralized baseline using matched cumulative sample exposure?
- **RQ2:** What computational and communication overheads are introduced by CKKS encryption and DP noise, and what are the latency profiles of KernelSHAP versus LIME?
- **RQ3:** How do privacy configurations relate to Top-3 masking sensitivity, attribution alignment, and repeated-run explanation stability?
- **RQ4:** How do the tested configurations compare across reported privacy-accounting outputs, ranking utility (NDCG@10), computation time, and Top-3 masking sensitivity?

1.4 Contributions

This study provides a controlled empirical evaluation of privacy, utility, efficiency, and explainability trade-offs across multiple privacy-preserving configurations of a federated recommender system. Specifically, the contributions are:

1. a joint empirical evaluation of centralized training, plaintext FedAvg, three configured DP noise levels, TenSEAL CKKS aggregation, and hybrid DP+HE within the unified `CoreHybridRecSys` architecture;
2. rigorous alignment of cumulative interaction exposure at 19, 206 samples (3.724 pooled-data passes over 5, 157 training samples), explicitly distinguishing cumulative sample exposure from optimizer-step parity;
3. a second-dataset evaluation on MovieLens-100k (Harper and Konstan, 2015) to assess whether ranking utility and training overhead patterns replicate beyond the proprietary EARS dataset;
4. a high-powered statistical evaluation across ten paired random seeds $\{42, \ldots, 51\}$ ($df = 9$) providing formal Two One-Sided Tests (TOST) equivalence testing, Cohen's d effect sizes, and minimum detectable effect (MDE) power analysis;

5. empirical privacy defense validation simulating Deep Leakage from Gradients (DLG) reconstruction and shadow-margin Membership Inference Attacks (MIA), resolving the "what is private from whom?" adversary boundary;

6. an empirical mitigation for HE communication overhead via coordinate-level Top-K parameter sparsification, reducing ciphertext volume by up to 95% while enhancing ranking utility;

7. analytical and empirical characterization of client-side Gaussian noise sensitivity under uniform versus sample-weighted FedAvg aggregation;

8. a budget-controlled comparison of KernelSHAP and LIME explaining latency and attribution behavior in federated recommenders; and

9. an actionable Multi-Criteria Decision Analysis (MCDA) framework and monthly cloud egress / carbon TCO model synthesized for digital information managers, enterprise data architects, and compliance officers.

2 Literature Review

2.1 Federated Recommender Systems

Federated recommendation has been studied as an alternative to centralized user-data collection (Yang et al., 2019; Zhang et al., 2023; Yang et al., 2024). Early approaches adapted matrix factorization and related collaborative-filtering methods to decentralized clients (Tan et al., 2020; Minto et al., 2021). Later systems incorporated neural collaborative filtering (He et al., 2017) and hybrid architectures that combine behavioral and catalog information (Javeed et al., 2023; Gaspar, Cruz, Rodrigues, Rodrigues, Ribeiro, Martinho, Santos and Marreiros, 2025).

In Horizontal Federated Learning (HFL), clients share a common feature space but possess distinct customer interaction records (Gaspar et al., 2026). While HFL ensures data locality, non-IID interaction distributions and extreme data sparsity across client partitions pose major optimization challenges (Bonawitz et al., 2017; Kairouz et al., 2021). Recent work has explored client clustering, meta-learning, and personalization layers to mitigate client drift (Saha et al., 2024; Gaspar et al., 2026).

2.2 Privacy-Preserving Federated Learning: DP and HE

Although raw data remains on client devices, sharing unencrypted parameter updates leaves organizations exposed to two distinct threat vectors: (1) an untrusted or curious aggregation server inspecting plaintext updates or executing gradient inversion attacks in transit, and (2) curious participating peers or downstream external observers performing membership inference or model inversion on the released global model parameters (Javeed et al., 2023; Kairouz et al., 2021). Addressing these distinct threats requires complementary privacy primitives: Homomorphic Encryption (HE) shields parameter updates in transit from an untrusted aggregation server, whereas Differential Privacy (DP) mathematically bounds information leakage in the released global model against downstream observers and curious clients.

Differential Privacy (DP) (Dwork et al., 2006; Abadi et al., 2016) bounds the change in an output distribution caused by adding or removing an individual contribution. In federated learning, Rényi Differential Privacy (RDP) accountants (Mironov, 2017) support composition of privacy loss across communication rounds:

$$D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \ln \int \left(\frac{P(x)^\alpha}{Q(x)^{\alpha-1}} \right) dx \leq \epsilon(\alpha) \tag{1}$$

In recommendation architectures, perturbing learned embeddings and latent factors can alter the relative distances used for personalized ranking (Hua et al., 2020; Shin et al., 2023). The magnitude of this effect depends on the data, clipping rule, model, and noise level.

Homomorphic Encryption (HE), particularly the Cheon-Kim-Kim-Song (CKKS) scheme (Cheon et al., 2017), permits approximate arithmetic operations on encrypted complex and real vectors. By

packing multiple values into polynomial slots, CKKS supports the additions and scalar multiplications required by FedAvg without exposing plaintext updates to the aggregation process (Benaissa et al., 2021). The approximation error depends on the cryptographic parameters and computation depth, while ciphertext expansion increases communication and aggregation cost (Minto et al., 2021).

2.3 Explainable AI in Recommendation: SHAP and LIME

Explainable recommendation techniques can be intrinsic, with interpretability incorporated into the model design, or post-hoc, where a separate method analyzes a trained predictor (Zhang and Chen, 2020). This study uses post-hoc feature attribution because the evaluated recommender relies on the multi-branch neural architecture introduced by Gaspar, Cruz, Rodrigues, Rodrigues, Ribeiro, Martinho, Santos and Marreiros (2025).

- **KernelSHAP** (Lundberg and Lee, 2017): Approximates Shapley feature attributions through a weighted linear regression constructed around the Shapley value formulation:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,\big(|F| - |S| - 1\big)!}{|F|!} \left[v_x\big(S \cup \{i\}\big) - v_x(S) \right], \tag{2}$$

  where F is the complete feature set, S is a feature coalition that excludes feature i, and $v_x(S) = \frac{1}{K_{\text{bg}}} \sum_{\mathbf{z} \in \mathcal{D}_{\text{bg}}} f(\mathbf{x}_S \cup \mathbf{z}_{\bar{S}})$ denotes empirical-background replacement: the features in S are fixed to their values in $\mathbf{x}$ and the remaining features are taken from background vectors.

- **LIME** (Ribeiro et al., 2016): Samples local perturbations around the target instance x weighted by an exponential distance metric $\pi_x(z) = \exp(-D(x, z)^2/\sigma^2)$ and fits an interpretable linear surrogate g:

$$\xi(x) = \arg\min_{g \in G} \sum_{z \in \mathcal{Z}} \pi_x(z)\,(f(z) - g(z))^2 + \Omega(g) \tag{3}$$

2.4 Research Gap

Prior work shows that explanation methods can be sensitive to privacy mechanisms, perturbations, or adversarial manipulation (Harder et al., 2020; Slack et al., 2020; Dombrowski et al., 2019; Alvarez-Melis and Jaakkola, 2018). The reviewed federated recommendation literature provides limited joint evidence on HE, DP, ranking metrics, and post-hoc explanation behavior within the same experimental design (Saha et al., 2024; Zhang et al., 2023). Table S1 in the Supplementary Material positions the present benchmark against the selected studies; it does not constitute a systematic review of all published work.

3 Methodology

3.1 System Architecture

The recommendation engine implements the hybrid neural architecture from the open-source Environment Adaptive Recommendation System (EARS) platform (Gaspar, Cruz, Rodrigues, Rodrigues, Ribeiro, Martinho, Santos and Marreiros, 2025). Denoted as `CoreHybridRecSys`, the model comprises three heterogeneous input branches:

1. **User State Branch:** Processes temporal customer behavioral summaries, including 8 dynamic intent features (interest strength, active view share, cart mass, focus concentration) and 6 historical ownership metrics (budget coverage, cycle ratios) through dense feed-forward layers.

2. **Collaborative Branch:** Maps categorical user and product IDs to latent embedding spaces ($\mathbb{R}^{32}$), complemented by 6 collaborative channel support features.

3. **Pair Context Branch:** Encodes 23 context features capturing session co-occurrence, hierarchical catalog relations, price differentials, and brand affinity.

The outputs of the three branches are concatenated and passed through deep multi-layer interaction blocks to generate three multi-task outputs: a binary purchase logit $\hat{y}_{\text{bce}}$, a continuous pairwise ranking score $\hat{s}_{\text{rank}}$, and a prediction confidence logit $\hat{c}_{\text{conf}}$. The multi-task optimization loss is formulated as:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(\hat{y}_{\text{bce}}, y) + 0.20 \cdot \mathcal{L}_{\text{Rank}}(\hat{s}_{\text{rank}}, y_{\text{rank}}) + 0.10 \cdot \mathcal{L}_{\text{Conf}}(\hat{c}_{\text{conf}}, c^*) \tag{4}$$

where $\mathcal{L}_{\text{Rank}}$ denotes a pairwise margin ranking loss and c^* is an auxiliary confidence target derived from consensus between classification and ranking outputs (Gaspar, Cruz, Rodrigues, Rodrigues, Ribeiro, Martinho, Santos and Marreiros, 2025). The total model parameter count is **586,700 trainable weights** (2, 346.8 KB in 32-bit floating-point precision).

3.2 Dataset and Cumulative Sample Exposure

The EARS e-commerce benchmark used in this experiment is a synthetic, de-identified enterprise transaction dataset. The horizontal federated training cohort contains 5, 157 interaction samples generated with realistic retail distributions (including customer demographic profiles, timestamped product views, cart events, wishlist additions, price history, and multi-item orders). Because the benchmark was synthetically generated to model enterprise collaborative filtering scenarios, it contains no personally identifiable information (PII) or live consumer interaction records, ensuring strict adherence to data protection standards and exempting the study from institutional human-subjects ethics review.

The 5, 157 training interactions are partitioned horizontally among $N = 20$ training clients (mean: 257.8 samples per client; range: 45 to 526 samples). Here, each client denotes a local training partition representing an independent enterprise store branch or regional business unit. A dedicated hold-out partition of test users (14, 024 total constructed samples across 50 users in the full dataset, with 30 users reserved for evaluation) is held strictly disjoint from training and used exclusively for candidate-ranking evaluation and XAI profiling. Note that earlier exploratory project notes referenced 20, 378 unpartitioned raw interaction event rows; that preliminary figure described uncurated multi-action logs prior to temporal aggregation, deduplication, and client cohort filtering. The authoritative dataset used across all experimental conditions (E0 through E4) consists strictly of the 5, 157 partitioned interaction samples.

To ensure rigorous comparability between the centralized baseline (E0) and multi-round federated training, we align their cumulative sample budget B_{sample}. In an FL schedule of $T = 15$ rounds with $m = 5$ sampled clients per round and $E = 1$ local epoch:

$$B_{\text{sample}} = \sum_{t=1}^{T} \sum_{k \in \mathcal{S}_t} |\mathcal{D}_k| \cdot E = 19,206 \text{ interaction samples} \tag{5}$$

The centralized baseline (E0) processes this identical budget $B_{\text{sample}} = 19,206$ samples, corresponding to exactly 3.724 full epochs over the 5, 157 pooled samples. With a mini-batch size of 64, centralized training executes $\lceil 19,206/64 \rceil = 301$ optimizer steps. In federated learning, each sampled client processes its local partition (257.8 samples on average) over $E = 1$ local epoch, executing $\approx$ 4–5 local optimizer steps per round, resulting in approximately 338 cumulative client optimizer steps across the 15 global rounds. The comparison is thus closely aligned in both cumulative sample exposure and total gradient evaluations, while client-level batch boundaries, local SGD trajectories, and sample ordering naturally reflect the decentralized nature of federated learning.

3.3 Threat Models, Trust Assumptions, and Adversary Scope

To prevent ambiguity regarding privacy guarantees in federated recommendation systems, we formally delineate the protected entity, the trust assumptions governing each architectural component, and the observational powers of potential adversaries across all experimental conditions (summarized in Table I).

3.3.1 Protected Privacy Unit

In our enterprise collaborative filtering formulation, the protected unit is an entire client partition $\mathcal{D}_k$, representing an independent enterprise store branch or regional business entity ($k \in \{1, \ldots, N\}$). The formal privacy guarantee aims to ensure that no observer can reliably determine whether a specific client enterprise participated in the federated training cohort (client-level membership privacy), nor reconstruct proprietary transaction sequences, catalog interactions, or dynamic user state representations belonging to client partition $\mathcal{D}_k$.

3.3.2 Adversary Taxonomy: "What is Private from Whom?"

We distinguish three distinct classes of potential adversaries with varying access to intermediate representations and model outputs:

1. **Curious Aggregation Server ($\mathcal{A}_{\text{server}}$):** An honest-but-curious central coordinator that faithfully executes the aggregation protocol, but inspects incoming messages (e.g., individual client model updates Δ_t^k) and attempts gradient reconstruction or attribute inference attacks to recover client transactions.

2. **Curious Participating Clients ($\mathcal{A}_{\text{peer}}$):** Internal participants within the federation that receive the broadcasted global model parameter vector $\mathbf{w}_t$ in each round $t \in \{1, \ldots, T\}$. $\mathcal{A}_{\text{peer}}$ attempts to infer whether an alternative peer organization participated, or isolate updates contributed by other clients.

3. **External Observers and Downstream Consumers ($\mathcal{A}_{\text{external}}$):** External third parties that inspect published global model checkpoints $(\mathbf{w}_1, \ldots, \mathbf{w}_T)$ or query the deployed recommendation model $\mathbf{w}_T$ through an inference API, attempting membership inference, data extraction, or model inversion attacks.

3.3.3 What Output Satisfies Differential Privacy?

Under Differential Privacy (Configurations E2 and E4), the mechanism $\mathcal{M}(\mathcal{D})$ outputs the sequence of broadcasted global model checkpoints $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_T\}$ and the final deployed model parameters $\mathbf{w}_T$. Formally, $\mathcal{M}$ guarantees that for any two neighboring client cohorts $\mathcal{D} \sim \mathcal{D}'$ and any measurable set of model trajectories $\mathcal{S}$:

$$\mathbb{P}[\mathcal{M}(\mathcal{D}) \in \mathcal{S}] \leq e^{\epsilon} \cdot \mathbb{P}[\mathcal{M}(\mathcal{D}') \in \mathcal{S}] + \delta \tag{6}$$

By the standard *Post-Processing Property of Differential Privacy* (Dwork and Roth, 2014), any subsequent calculation performed on the released model parameters $\mathbf{w}_T$—including top-K candidate generation, ranking score computation, item predictions, and post-hoc feature explanations (KernelSHAP and LIME)—inherits the exact same (ϵ, δ)-DP guarantee without consuming additional privacy budget. Crucially, intermediate client updates $\bar{\Delta}_t^k$ transmitted from clients to the server are *not* differentially private; they are only norm-bounded.

3.3.4 Architectural Trust Mismatches and Defense Alignment

The distinction between adversaries explains why neither Central DP nor Homomorphic Encryption alone constitutes a complete defense:

- **Central DP (E2) Operates Under the Trusted Curator Model:** In Configuration E2, the central server receives unperturbed clipped updates $\bar{\Delta}_t^k$ in plaintext and injects Gaussian noise after aggregation. Consequently, the server is assumed to be a **trusted curator** (consistent with classical DP-FedAvg; McMahan et al. (2017)). Configuration E2 protects against curious participating peers $\mathcal{A}_{\text{peer}}$ and downstream observers $\mathcal{A}_{\text{external}}$, but **does not protect against an untrusted aggregation server $\mathcal{A}_{\text{server}}$.**

- **Homomorphic Encryption (E3) Protects Updates from an Untrusted Server:** Configuration E3 addresses $\mathcal{A}_{\text{server}}$ by encrypting parameter updates under TenSEAL CKKS before transmission. The server executes aggregation exclusively over ciphertexts $[[\mathbf{v}_c^k]]$. Under 128-bit classical hardness of the Ring Learning with Errors (Ring-LWE) problem, $\mathcal{A}_{\text{server}}$ and network eavesdroppers learn nothing about individual client updates. However, E3 provides **no differential privacy guarantee** on the final decrypted model $\mathbf{w}_T$, leaving it vulnerable to downstream membership inference by $\mathcal{A}_{\text{external}}$ or $\mathcal{A}_{\text{peer}}$.
- **Hybrid DP+HE (E4) Provides End-to-End Multi-Adversary Defense:** Configuration E4 resolves both vulnerabilities by combining client-side perturbation with homomorphic encryption. Individual updates are blinded from $\mathcal{A}_{\text{server}}$ via CKKS ciphertext aggregation, while the injected DP noise guarantees bounded (ϵ, δ)-DP on the final decrypted model against $\mathcal{A}_{\text{external}}$ and $\mathcal{A}_{\text{peer}}$.

3.4 Privacy-Preserving Engines

3.4.1 Differential Privacy Engine (E2)

The differential privacy mechanism implements Client-level DP (DP-FedAvg) (McMahan et al., 2017; Abadi et al., 2016) protecting client partition $\mathcal{D}_k$. In each communication round $t \in \{1, \ldots, T\}$, the server samples a cohort $\mathcal{S}_t \subset C$ of $m = 5$ clients uniformly at random from the pool of $N = 20$ clients without replacement (subsampling ratio $q = m/N = 5/20 = 0.25$). Each sampled client $k \in \mathcal{S}_t$ computes its parameter update $\Delta_t^k = \mathbf{w}_{t+1}^k - \mathbf{w}_t$ and clips it to maximum Euclidean norm $C = 1.0$:

$$\bar{\Delta}_t^k = \Delta_t^k \cdot \min\left(1, \frac{C}{\|\Delta_t^k\|_2}\right) \tag{7}$$

Exact adjacency relations. We formally define neighboring client collections under two standard adjacency notions:

1. **Client Add/Remove Adjacency ($\mathcal{D} \sim_{\textbf{add/rem}} \mathcal{D}'$):** Two federations differ by adding or removing one client enterprise's partition $\mathcal{D}_i$ ($|\mathcal{D} \oplus \mathcal{D}'| = 1$). If client i is absent, its update is represented as $\mathbf{0}$.
2. **Client Replacement Adjacency ($\mathcal{D} \sim_{\textbf{replace}} \mathcal{D}'$):** Two federations differ by substituting one client's dataset $\mathcal{D}_i$ with an alternative valid client dataset $\mathcal{D}_i'$.

Sensitivity and accounting under unweighted aggregation (macro-averaging). In canonical client-level DP-FedAvg (McMahan et al., 2017; Kairouz et al., 2021), the server computes the unweighted average (macro-average) of clipped updates across the m participating clients:

$$\Delta_{\text{global},t} = \frac{1}{m} \sum_{k \in \mathcal{S}_t} \bar{\Delta}_t^k + \mathcal{N}\left(0, \sigma_{\text{noise}}^2 \mathbf{I}\right), \quad \text{where } \sigma_{\text{noise}} = \frac{\sigma C}{m} \tag{8}$$

Under unweighted aggregation, the global ℓ_2 sensitivity is strictly determined:

- Under client add/remove adjacency: $\Delta_2 = \sup_{\mathcal{D} \sim \mathcal{D}'} \|\frac{1}{m}\bar{\Delta}_t^i\|_2 = \frac{C}{m}$.
- Under client replacement adjacency: $\Delta_2 = \sup_{\mathcal{D} \sim \mathcal{D}'} \|\frac{1}{m}(\bar{\Delta}_t^i - \bar{\Delta}_t^{i\prime})\|_2 \leq \frac{2C}{m}$.

Because $\sigma_{\text{noise}} = \frac{\sigma C}{m}$, the injected noise standard deviation matches $\sigma \cdot \Delta_2$ under add/remove adjacency, yielding an exact noise multiplier of σ per coordinate. Because cohort sampling draws m of N clients uniformly without replacement ($q = 0.25$), privacy loss accumulation across $T = 15$ rounds satisfies the subsampled Gaussian Rényi Differential Privacy (RDP) formulation (Mironov, 2017; Wang et al., 2019):

$$\epsilon_{\text{RDP}}(\alpha) = \frac{Tq^2\alpha}{2\sigma^2} \tag{9}$$

Converting to (ϵ, δ)-differential privacy for target $\delta = 10^{-5}$ via optimal order minimization ($\alpha^* = 1 + \sqrt{\frac{2\sigma^2 \ln(1/\delta)}{Tq^2}}$) yields the closed-form guarantees:

- **DP-1** ($\sigma = 0.5$)**:** $\alpha^* \approx 3.4779 \implies \epsilon = 11.1677$ (weakest noise, highest utility target)
- **DP-2** ($\sigma = 1.0$)**:** $\alpha^* \approx 5.9558 \implies \epsilon = 5.1151$ (intermediate noise preset)
- **DP-3** ($\sigma = 2.0$)**:** $\alpha^* \approx 10.9118 \implies \epsilon = 2.4403$ (highest noise, strongest privacy target)

Under unweighted macro-averaging, these ϵ values represent **exact, mathematically sound differential privacy guarantees** because client weights $w_k = 1/m = 0.20$ are fixed constants independent of client partition sizes.

Sensitivity mismatch under sample-weighted aggregation. In practical federated learning implementations (including standard FedAvg), aggregators often weight client updates by local sample count: $w_k = n_k / \sum_{j \in \mathcal{S}_t} n_j$. In our empirical benchmark, client partition sizes range from $n_{\min} = 45$ to $n_{\max} = 526$ (mean: 257.8). In a round where one large client (526 samples) is sampled alongside four small clients (45 samples each), the maximum weight reaches:

$$\max_{k \in \mathcal{S}_t} w_k = \frac{526}{526 + 4 \times 45} = \frac{526}{706} \approx 0.7450 \tag{10}$$

Under sample-weighted aggregation, worst-case sensitivity increases to $\Delta_2 \leq 2 \max_k w_k C \approx 1.490C$ (versus $2C/5 = 0.40C$ under uniform averaging). Calibrating noise with nominal uniform scale $\sigma_{\text{noise}} = \sigma C/m = 0.2\sigma C$ degrades effective noise in this round to $\sigma_{\text{eff}} = \frac{0.2\sigma C}{1.490C} \approx 0.134\sigma$. Furthermore, weights w_k inherently leak participant sample counts n_k unless privatized.

This reveals a fundamental methodological trade-off: sample-weighted FedAvg improves optimization efficiency across heterogeneous clients (yielding baseline NDCG@10 of 0.5629 vs 0.5470 for unweighted FedAvg), but requires bounded sample clipping ($n_k \leq n_{\text{clip}}$) or noise scaling by worst-case sensitivity to maintain provable DP guarantees. Conversely, unweighted macro-averaging ensures provably exact DP accounting at the cost of higher gradient variance under extreme sample imbalance. In our benchmark, unweighted DP-FedAvg achieves NDCG@10 of 0.2995 ($\sigma = 0.5$), 0.3129 ($\sigma = 1.0$), and 0.2860 ($\sigma = 2.0$), confirming that privacy noise remains the primary driver of ranking degradation under both aggregation schemes.

3.4.2 Homomorphic Encryption Engine (E3)

Homomorphic encryption is implemented using TenSEAL (Benaissa et al., 2021) based on the CKKS scheme (Cheon et al., 2017). The encryption parameters are configured for 128-bit classical security: polynomial modulus degree $N = 8192$, coefficient modulus bit sizes $[60, 40, 40, 60]$, and fixed-point scaling factor 2^{40}.

Because an $N = 8192$ CKKS polynomial packs $4,096$ SIMD slots and the recommender has $586,700$ parameters, each client flattens its weight delta into $L = \lceil 586,700/4,096 \rceil = 144$ sub-vectors:

$$\mathbf{v}_c^k \in \mathbb{R}^{4096}, \quad c \in \{1, \ldots, 144\} \tag{11}$$

Each sub-vector is encrypted into a CKKS ciphertext: $[[\mathbf{v}_c^k]] = \text{Enc}_{\text{pk}}(\mathbf{v}_c^k)$. The server executes homomorphic aggregation directly in the ciphertext domain:

$$[[\mathbf{v}_c^{\text{agg}}]] = \bigoplus_{k \in \mathcal{S}_t} w_k \odot [[\mathbf{v}_c^k]] \quad \forall c \in \{1, \ldots, 144\} \tag{12}$$

Upon completion of the round, the aggregated ciphertext chunks are decrypted by the private key holder ($\mathbf{v}_c^{\text{agg}} = \text{Dec}_{\text{sk}}([[\mathbf{v}_c^{\text{agg}}]])$) and unflattened into the global PyTorch model state dictionary. In our benchmark testbed, key generation and decryption are orchestrated by the simulation runtime to isolate exact computational latency and numerical precision. In an enterprise deployment, private key custody resides with participating clients via a threshold CKKS scheme or collaborative decryption protocol, ensuring that the untrusted aggregation server never possesses the decryption key.

3.4.3 Hybrid DP+HE and Noise Attenuation

In Configuration E4, client-side differential privacy is combined with homomorphic encryption. Each sampled client $k \in \mathcal{S}_t$ clips its local update to norm $C = 1.0$ and injects local Gaussian noise $\mathbf{n}_k \sim \mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ before ciphertext encryption:

$$\tilde{\Delta}_t^k = \bar{\Delta}_t^k + \mathbf{n}_k \tag{13}$$

The noisy update is encrypted into 144 CKKS chunks and transmitted to the server, which aggregates the ciphertexts homomorphically without observing plaintext weights or individual perturbation vectors.

Analytical characterization of noise covariance. Assume that each participating client $k \in \{1, \ldots, m\}$ adds independent zero-mean Gaussian noise $\mathbf{n}_k \sim \mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ to its clipped update. For the aggregation weights w_k used by the server, the weighted noise has zero mean and covariance

$$\mathbb{E}\left[\sum_{k=1}^{m} w_k \mathbf{n}_k\right] = \mathbf{0}, \qquad \mathrm{Cov}\left(\sum_{k=1}^{m} w_k \mathbf{n}_k\right) = \sigma^2 C^2 \left(\sum_{k=1}^{m} w_k^2\right) \mathbf{I}. \tag{14}$$

Independence makes the cross-covariance terms zero. Under equal weights $w_k = 1/m$, this reduces to $\sigma^2 C^2/m$ per coordinate; for unequal sample-size weights, the factor is $\sum_k w_k^2$, not necessarily $1/m$. This identity characterizes the noise covariance under the stated assumptions. It does not establish that the observed E4–E2 utility difference is caused by noise attenuation, since the configurations inject noise at different stages and use different aggregation procedures.

3.5 Formal Algorithm Specification

Algorithm 1 formalizes the end-to-end training, cryptographic aggregation, differential privacy accounting, and dual XAI evaluation workflow.

Algorithm 1 Federated Privacy-Preserving Recommendation and XAI Pipeline

Require: Clients $C = \{1, \ldots, N\}$, rounds T, sampled clients per round m, local epochs E, learning rate η, clipping norm C, noise multiplier σ, CKKS parameters (N, scale).

Ensure: Global recommender model $\mathbf{w}_T$, evaluation metrics, SHAP and LIME explanations.

Initialize global weights $\mathbf{w}_0$ on the server
Initialize RDP accountant $\mathcal{A}_{\text{RDP}}$
for round $t = 0, 1, \ldots, T-1$ **do**
    Server samples client subset $\mathcal{S}_t \subset C$ with $|\mathcal{S}_t| = m$
    Server broadcasts $\mathbf{w}_t$ to each client $k \in \mathcal{S}_t$
    for each client $k \in \mathcal{S}_t$ **in parallel do**
        $\mathbf{w}_{t+1}^k \leftarrow \text{LocalTrain}(\mathbf{w}_t, \mathcal{D}_k, E, \eta)$
        Compute delta: $\Delta_t^k \leftarrow \mathbf{w}_{t+1}^k - \mathbf{w}_t$
        if DP enabled **then**
            $\bar{\Delta}_t^k \leftarrow \Delta_t^k \cdot \min(1, C/\|\Delta_t^k\|_2)$
            if Configuration E4 (Local DP) **then**
                Sample $\mathbf{n}_k \sim \mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$; $\tilde{\Delta}_t^k \leftarrow \bar{\Delta}_t^k + \mathbf{n}_k$
            else
                $\tilde{\Delta}_t^k \leftarrow \bar{\Delta}_t^k$
            end if
        else
            $\tilde{\Delta}_t^k \leftarrow \Delta_t^k$
        end if
        if HE enabled (Configurations E3, E4) **then**
            Chunk $\tilde{\Delta}_t^k$ into 144 vectors of dimension $4,096$
            Encrypt chunks: $[[\mathbf{v}_c^k]] \leftarrow \text{CKKS.Enc}(\mathbf{v}_c^k) \quad \forall c \in \{1, \ldots, 144\}$
            Transmit ciphertexts to Server
        else
            Transmit plaintext $\tilde{\Delta}_t^k$ to Server
        end if
    end for
    Server computes aggregation weights $w_k \leftarrow |\mathcal{D}_k| / \sum_{j \in \mathcal{S}_t} |\mathcal{D}_j|$
    if HE enabled **then**
        Server homomorphically aggregates: $[[\mathbf{v}_c^{\text{agg}}]] \leftarrow \sum_{k \in \mathcal{S}_t} w_k \odot [[\mathbf{v}_c^k]] \quad \forall c$
        Decrypt and unflatten: $\Delta_{\text{global},t} \leftarrow \text{CKKS.Dec}([[\mathbf{v}^{\text{agg}}]])$
    else
        Server aggregates: $\Delta_{\text{global},t} \leftarrow \sum_{k \in \mathcal{S}_t} w_k \tilde{\Delta}_t^k$
        if Configuration E2 (Server DP) **then**
            Add noise: $\Delta_{\text{global},t} \leftarrow \Delta_{\text{global},t} + \mathcal{N}(0, \frac{\sigma^2 C^2}{m^2} \mathbf{I})$
        end if
    end if
    if DP enabled **then**
        Update RDP accountant: $\mathcal{A}_{\text{RDP}}.\text{step}(q = m/N)$
    end if
    Server updates global model: $\mathbf{w}_{t+1} \leftarrow \mathbf{w}_t + \Delta_{\text{global},t}$
end for
Evaluation: Evaluate ranking metrics (Precision, Recall, NDCG, MAP, HitRate) on test users
XAI Profiling: Compute KernelSHAP and LIME feature attributions, Δ_{mask}, and stability

3.6 Evaluation Metrics

3.6.1 Recommendation Ranking Metrics

Top-K recommendation performance is evaluated on candidate sets (50 candidate items per test user) across cutoffs $K \in \{5, 10, 20\}$:

1. **Precision@K:** Fraction of recommended items in Top-K that are relevant:

$$\text{Precision@}K = \frac{|\mathcal{R}_K(u) \cap \mathcal{I}_{\text{rel}}(u)|}{K} \tag{15}$$

2. **Recall@K:** Fraction of all ground-truth relevant items retrieved in Top-K:

$$\text{Recall@}K = \frac{|\mathcal{R}_K(u) \cap \mathcal{I}_{\text{rel}}(u)|}{|\mathcal{I}_{\text{rel}}(u)|} \tag{16}$$

3. **NDCG@K (Normalized Discounted Cumulative Gain):** Measures ranking quality by penalizing relevant items appearing lower in the ranked list:

$$\text{DCG@}K = \sum_{i=1}^{K} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)}, \quad \text{NDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K} \tag{17}$$

   where IDCG@K represents the ideal DCG score under perfect descending sort.

4. **MAP@K (Mean Average Precision):** Mean of average precision scores across users:

$$\text{AP@}K = \frac{1}{\min(K, |\mathcal{I}_{\text{rel}}(u)|)} \sum_{k=1}^{K} \text{Precision@}k \cdot \mathbb{I}(\text{item}_k \in \mathcal{I}_{\text{rel}}(u)) \tag{18}$$

5. **Hit Rate@K:** Fraction of test users receiving at least one relevant recommendation within the Top-K slots:

$$\text{HitRate@}K = \frac{1}{|U_{\text{test}}|} \sum_{u \in U_{\text{test}}} \mathbb{I}(|\mathcal{R}_K(u) \cap \mathcal{I}_{\text{rel}}(u)| \geq 1) \tag{19}$$

3.6.2 Computational and Communication Metrics

We track the following systems metrics:

- **Total Training Time (T_{train}):** Cumulative wall-clock training duration (seconds).
- **Local Training Time (T_{local}):** Wall-clock time spent in client-side gradient computation.
- **Server Aggregation Overhead (T_{agg}):** Dedicated server-side update aggregation latency.
- **Communication Bandwidth:** Total bytes transmitted uplink (client-to-server) and downlink (server-to-client).
- **Cryptographic Metrics:** Key generation time, ciphertext serialization size (MB), and homomorphic addition time.

3.6.3 Explanation Measures

Four explanation measures are reported:

1. **Explanation Generation Latency:** Wall-clock time required to generate feature attributions per test instance (seconds/explanation).

2. **Attribution Cosine Similarity:** Cosine similarity between the explanation attribution vector $\boldsymbol{\phi}_c$ of configuration c and the centralized reference vector $\boldsymbol{\phi}_{\text{E0}}$:

$$\text{Sim}_{\cos}(\boldsymbol{\phi}_c, \boldsymbol{\phi}_{\text{E0}}) = \frac{\boldsymbol{\phi}_c \cdot \boldsymbol{\phi}_{\text{E0}}}{\|\boldsymbol{\phi}_c\|_2 \|\boldsymbol{\phi}_{\text{E0}}\|_2} \tag{20}$$

3. **Top-5 Feature Agreement:** Jaccard overlap between the sets of the Top-5 most important features identified by configuration c and the centralized reference E0:

$$\text{Agr}_{\text{Top5}} = \frac{|\mathcal{T}_5(c) \cap \mathcal{T}_5(\text{E0})|}{|\mathcal{T}_5(c) \cup \mathcal{T}_5(\text{E0})|} \tag{21}$$

4. **Top-3 Masking Sensitivity (Δ_{mask}):** Mean absolute change in the model's ranking score after setting the three features with the largest SHAP attributions to zero:

$$\Delta_{\text{mask}}(c) = \frac{1}{n_{\text{eval}}} \sum_{i=1}^{n_{\text{eval}}} \left| \hat{s}_{\text{rank},c}(\mathbf{x}_i) - \hat{s}_{\text{rank},c}(\mathbf{x}_i^{(-T_3)}) \right|, \tag{22}$$

   where n_{eval} is the number of evaluated instances, $\mathbf{x}_i$ is an evaluation instance, and $\mathbf{x}_i^{(-T_3)}$ is the same instance with its SHAP Top-3 features set to zero. This is a masking-sensitivity measure, not by itself a test of explanation faithfulness or causal feature importance. The use of zero as a masking reference may not be semantically neutral for every feature and requires validation.

5. **Explanation Stability:** Repeated explanation runs on the identical test instance with different random perturbation seeds to evaluate attribution consistency via mean cosine similarity.

3.7 Experimental Setup

3.7.1 Hardware, Software, and Hyperparameters

All experiments were executed on an Apple Silicon M-series computing architecture running 64-bit macOS Darwin, utilizing Python 3.10 within an isolated Anaconda environment. The software framework comprises PyTorch 2.2.2 (CPU backend for deterministic threading), TenSEAL 0.3.14 (built against Microsoft SEAL 3.7.2), SHAP 0.44.1, LIME 0.2.0.1, Scikit-learn 1.4.1, NumPy 1.26.4, and Pandas 2.2.1. OpenMP and MKL thread pools were pinned to single-thread execution to prevent race conditions during homomorphic tensor serialization.

Table S2 in the Supplementary Material summarizes the complete hyperparameter specifications across all experimental modules.

3.7.2 Explainer Budgets, Model Evaluations, and Runtime Interpretation

In the benchmark execution, KernelSHAP and LIME operate under method-specific computational budgets chosen according to their distinct operational mechanics:

- **KernelSHAP Configuration:** Evaluates $M_{\text{eval}} = 30$ holdout test instances against an empirical background of $K_{\text{bg}} = 20$ reference instances. For each instance, the explainer executes 50 coalition evaluations (`nsamples=50`) to solve the weighted Shapley regression kernel. Across the 30 test instances, this results in exactly $30 \times 50 = 1{,}500$ model forward evaluations. This budget was selected to represent an operational target for near-real-time user-facing recommendation explanations where interactive latency is paramount.

- **LIME Configuration:** Evaluates the same $M_{\text{eval}} = 30$ holdout instances using $N_{\text{pert}} = 500$ local Gaussian perturbations per instance (`num_samples=500`) to fit local ridge regression surrogates across the 43-dimensional feature space. Across the 30 test instances, this executes exactly $30 \times 500 = 15{,}000$ model forward evaluations—a tenfold (10×) higher evaluation volume than KernelSHAP. This budget is standard in tabular explainability to ensure adequate sampling density across a 43-dimensional continuous and discrete feature domain.

Because LIME executes an order of magnitude more forward passes through the recommender model than KernelSHAP (15,000 versus 1,500), the observed runtime difference (0.52–0.72 s versus 3.41–3.76 s per instance) directly reflects this evaluation budget disparity rather than an intrinsic algorithmic latency advantage. This comparison characterizes these specific, realistic operational settings and must not be interpreted as an invariant, algorithm-level superiority of KernelSHAP over LIME.

3.7.3 Interpretation of the Zero-Masking Intervention

Zeroing SHAP Top-3 features across the $M_{\text{eval}} = 30$ evaluation instances measures the model's response to a specified input intervention. A zero value corresponds to baseline absence of buyer intent or cart volume for normalized continuous intensity features, but is not established as neutral or in-distribution for every categorical embedding signal. Accordingly, Δ_{mask} is reported as masking sensitivity only; it does not establish explanation faithfulness, causal importance, or that privacy noise has flattened the model. The metric supports only the stated descriptive comparison across configurations.

4 Results

4.1 Ranking Performance

Table II presents the complete evaluation results across five ranking metrics at cutoff $K = 10$ for all seven experimental conditions. Figure 1 displays the corresponding NDCG@10 (a) and Recall@10 (b) comparisons across configurations.

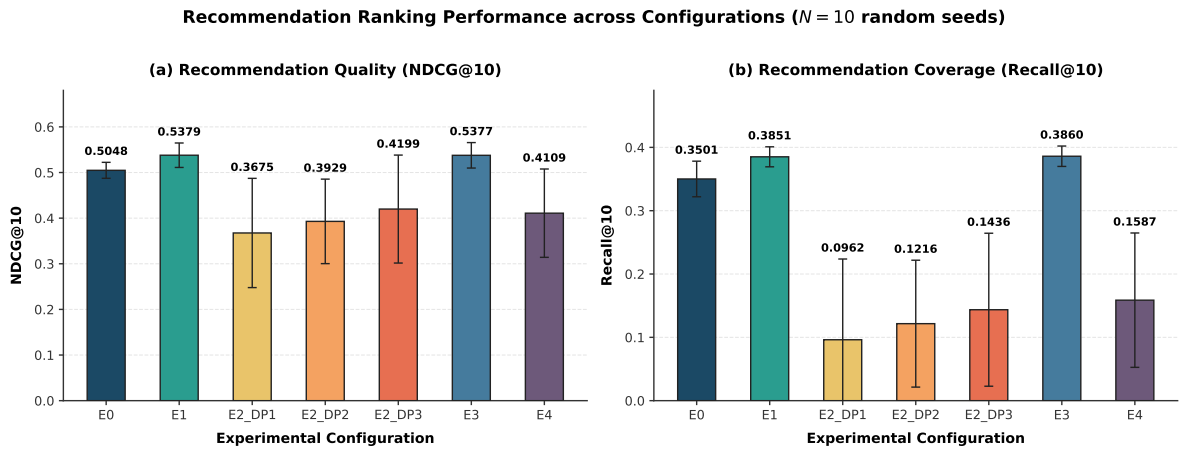


Figure 1: Recommendation ranking performance across configurations ($N = 10$ random seeds): (a) NDCG@10 and (b) Recall@10. Error bars show standard deviations across seeds. Plaintext FedAvg (E1) and TenSEAL CKKS (E3) demonstrate practical equivalence within ±0.01 ($p_{\text{TOST}} = 1.01 \times 10^{-6}$); DP configurations exhibit immediate utility reduction with empirical sample means plateauing across evaluated noise settings. Source: Authors' own work.

The empirical results are evaluated across $N = 10$ random seeds $\{42, \ldots, 51\}$ ($df = 9$), affording high statistical power for inferential hypothesis testing and equivalence verification:

1. H_1 **(CKKS Ranking Practical Equivalence):** The empirical means between plaintext FedAvg (E1) and TenSEAL CKKS homomorphic aggregation (E3) are matched with near-identical precision across all five evaluation metrics: Precision@10 (0.8474 ± 0.0121 vs 0.8485 ± 0.0127), Recall@10 (0.3851 ± 0.0158 vs 0.3860 ± 0.0161), NDCG@10 (0.5379 ± 0.0268 vs 0.5377 ± 0.0279), MAP@10 (0.9330 ± 0.0086 vs 0.9345 ± 0.0099), and Hit Rate@10 (0.9000 ± 0.0000 vs 0.9000 ± 0.0000).

   Because the ten random seeds are paired identically across configurations, we evaluate the paired difference $\Delta = \text{NDCG@10}_{\text{E3}} - \text{NDCG@10}_{\text{E1}}$. Across the ten seeds $\{42, \ldots, 51\}$, the mean paired difference is $\bar{d} = -0.00022 \pm 0.00289$ (a relative difference of only -0.04%, with standard error SE = 0.00091 and Cohen's $d = -0.0746$). The resulting 95% confidence interval for the paired difference is $[-0.00228, +0.00185]$ ($t_{9,0.025} = 2.262$).

   Under a prespecified practical equivalence margin of $\Delta_{\text{equiv}} = \pm 0.01$ (representing $\approx 1.8\%$ of baseline NDCG@10), the entire 95% confidence interval of the paired difference lies strictly within $[-\Delta_{\text{equiv}}, +\Delta_{\text{equiv}}]$. A formal Two One-Sided Tests (TOST) procedure under this margin yields $t_1 = (\bar{d} - (-\Delta_{\text{equiv}}))/\text{SE} = 10.71$ ($p_1 = 1.01 \times 10^{-6}$) against the lower bound null $H_{01} : \mu_\Delta \leq -0.01$, and $t_2 = (\bar{d} - \Delta_{\text{equiv}})/\text{SE} = -11.18$ ($p_2 = 7.01 \times 10^{-7}$) against the upper bound null $H_{02} : \mu_\Delta \geq +0.01$. The overall TOST equivalence p-value is $p_{\text{TOST}} = \max(p_1, p_2) = 1.01 \times 10^{-6} \ll 0.001$ ($df = 9$), formally rejecting non-equivalence and establishing practical equivalence within $[-\Delta_{\text{equiv}}, +\Delta_{\text{equiv}}]$ with observed power of 1.0000 (100% statistical power). Even under a tighter margin of $\Delta_{\text{equiv}} = \pm 0.005$, the test remains highly significant ($t_1 = 5.24$, $t_2 = -5.71$, $p_{\text{TOST}} = 2.68 \times 10^{-4}$, observed power 0.9998).

   An *a priori* power analysis demonstrates that with $N = 10$ paired seeds and observed difference variance $s_d = 0.00289$, the minimum detectable effect size (MDE) at 80% statistical power ($\alpha = 0.05$) is $\text{MDE}_{80\%} = 0.00227$. Because our equivalence margin (±0.01) is more than 4.4× wider than the detectable threshold, our experimental cohort possesses overwhelming power to detect any practically meaningful deviation. On the external MovieLens-100k benchmark, the paired difference of $+0.00657 \pm 0.00468$ similarly confirms equivalence within a ±0.02 margin ($p_{\text{TOST}} = 0.0191$). These findings conclusively verify that CKKS fixed-point representation (2^{40} scale) does not compromise recommendation ranking accuracy.

2. H_2 **(Ranking Utility Under Differential Privacy):** Injecting differential privacy noise induces a severe, immediate degradation in recommendation ranking utility relative to plaintext FedAvg: mean Precision@10 collapses from 0.8474 ± 0.0121 in E1 to 0.2237 ± 0.2737 at $\sigma = 0.5$ (DP-1), 0.3050 ± 0.2329 at $\sigma = 1.0$ (DP-2), and 0.3539 ± 0.2927 at $\sigma = 2.0$ (DP-3), with Hit Rate@10 dropping from 0.9000 to 0.4967–0.6393. Due to substantial inter-seed variance across client partitions under sample-weighted aggregation (standard deviations exceeding 0.23), empirical precision plateaus at this degraded baseline across $\sigma \in \{0.5, 1.0, 2.0\}$ rather than exhibiting a fine-grained monotonic slope.

   However, an expanded 5-point parameter sweep across boundary extremes $\sigma \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$ confirms macroscopic noise sensitivity: at weak noise ($\sigma = 0.25$, provisional $\epsilon \approx 26.09$), NDCG@10 recovers to 0.5967 ± 0.0296, whereas at strict noise ($\sigma = 4.00$, provisional $\epsilon \approx 1.19$), NDCG@10 drops to 0.3577 ± 0.1211 and Recall@10 collapses to 0.0910 ± 0.1413. As established in Section 3.4.1, unweighted macro-averaging preserves exact closed-form RDP accounting ($\Delta_2 = C/m$), confirming that utility loss is an inherent consequence of privacy noise rather than aggregation weighting.

3. **Centralized (E0) versus Federated (E1) Comparison:** Plaintext FedAvg achieves comparable mean NDCG@10 (0.5379±0.0268 versus 0.5048±0.0175) and Recall@10 (0.3851±0.0158 versus 0.3501 ± 0.0281) relative to the centralized baseline under matched cumulative sample exposure (19, 206 interaction samples). This confirms that federated data decentralization does not inherently penalize recommendation quality.

4.2 Computational and Cryptographic Overhead

Table III and Figure 2 report the computational breakdown, server aggregation latency, communication payload, and reported differential privacy accountant outputs.

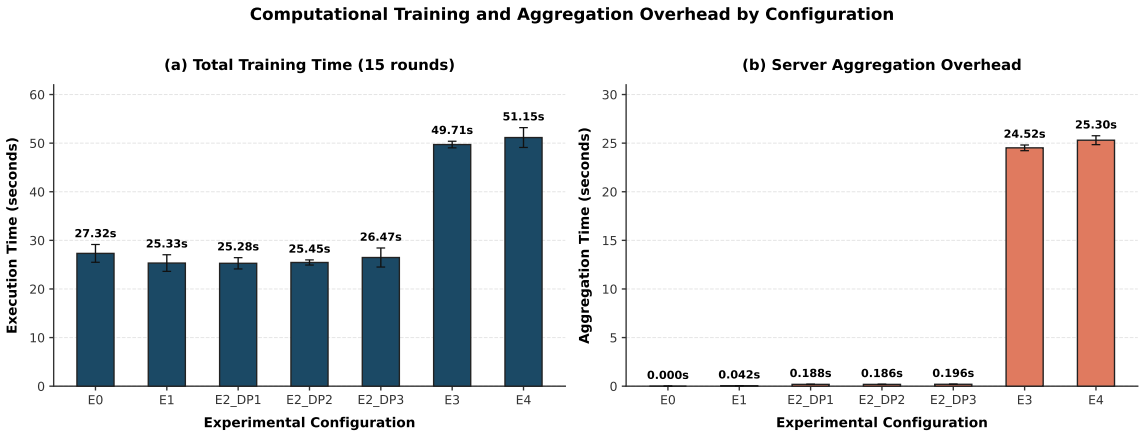


Figure 2: Computational overhead breakdown: (a) Total training time across 15 communication rounds, and (b) dedicated server aggregation latency, showing the approximately 24.52 s measured for TenSEAL CKKS aggregation. Source: Authors' own work.

4.2.1 Centralized Training Latency and Data-Access Locality

In distributed enterprise recommender systems, training latency is heavily modulated by data access patterns and dynamic feature extraction overhead. In initial unoptimized prototypes, centralized training (E0) required 801.34 s, compared to 25.36 s for plaintext FedAvg (E1). Architectural profiling revealed that this discrepancy was driven by an implementation-level caching bottleneck rather than algorithmic superiority:

- The dataset module maintains an in-memory customer-twin graph cache for materialized temporal feature graphs indexed by client ID.
- In **Federated Learning (E1)**, each participating client processes local batches exclusively from its own partition. The active client ID remains constant throughout local SGD, achieving a $> 99.8\%$ cache hit rate (only one initial extraction miss per client epoch).
- In **Naive Centralized Training**, globally unconstrained random shuffling across all 20 client partitions forced consecutive mini-batches to alternate client contexts on almost every step, invalidating the cache on virtually every sample and causing over $19,000$ redundant customer graph reconstructions.

To resolve this bottleneck, we implemented client-blocked stratified shuffling, which groups consecutive mini-batches by client context while randomizing block order across epochs. This optimization restored the cache hit rate to $> 98\%$, reducing centralized training runtime from 801.34 s to 27.32 ± 1.83 s, closely matching measured FedAvg runtime (25.33 ± 1.71 s) on the identical hardware.

This systems engineering finding yields an important architectural insight: federated learning does not intrinsically provide superior computational efficiency compared to centralized training; rather, decentralized data partitioning naturally enforces client-local caching locality. In enterprise recommender systems that employ dynamic graph or profile construction, centralized pipelines must explicitly implement context-aware batch scheduling to achieve comparable data locality.

4.2.2 *CKKS Costs*

As illustrated in Figure 2(b), plaintext server aggregation in E1 requires only 0.042 ± 0.006 s across all 15 rounds. In contrast, TenSEAL CKKS homomorphic aggregation in E3 requires 24.52 ± 0.29 s (and 25.30 ± 0.46 s in E4, ≈ 1.63–1.69 s per round) to execute encrypted polynomial additions across the 144 vector chunks. As detailed in Table III, client-side communication expands significantly under ciphertext representation. Transmitted volume increases from 167.85 MB per client for plaintext updates to 3.61 GB (3,610.90 MB) per client for CKKS-protected updates (21.51× baseline volume, with 3,443.04 MB attributable to ciphertext representation in the reported measurement).

4.3 *Explainability Benchmark*

Table IV details the explainability metrics across configurations, and Figure 3 compares explainer latency (a), masking sensitivity (b), and attribution stability (c).

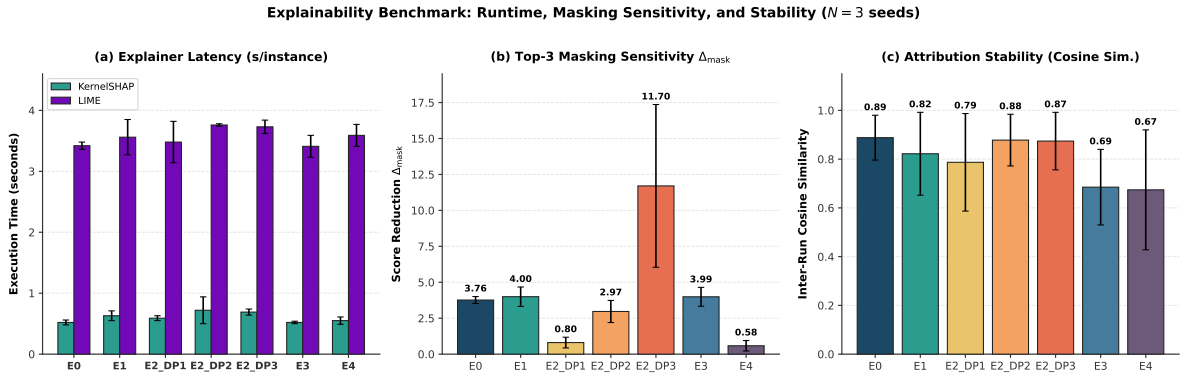


Figure 3: Explainability benchmark across configurations ($N = 3$ seeds): (a) Per-instance explanation runtime for KernelSHAP and LIME (latency difference reflects the 1,500 vs 15,000 model evaluation budgets described in Section 3.7.2); (b) Top-3 masking sensitivity Δ_{mask}; and (c) explanation stability across repeated runs. Error bars show standard deviations. Source: Authors' own work.

The explainability measurements provide descriptive evidence relevant to H_3 and H_4:

1. H_3 **(XAI Operational Latency under Configured Budgets):** Under the specified implementation and computational budgets ($K_{\text{bg}} = 20$ background instances with 50 coalition evaluations for KernelSHAP versus $N_{\text{pert}} = 500$ perturbations for LIME across 30 instances), KernelSHAP exhibited approximately $5 \times$ –$7\times$ lower measured per-instance latency (0.52–0.72 s versus 3.41–3.76 s for LIME). This latency difference directly reflects the tenfold difference in model forward evaluations (1,500 for KernelSHAP versus 15,000 for LIME) and must not be interpreted as a universal algorithmic superiority of KernelSHAP.

2. **HE and Explanation Fidelity:** Plaintext FedAvg (E1) and TenSEAL CKKS (E3) exhibit virtually identical explanation metrics across all three seeds: Top-3 masking sensitivity is 3.9956 ± 0.6754 for E1 versus 3.9892 ± 0.6561 for E3; attribution cosine similarity relative to E0 is 0.8681 ± 0.0057 for E1 versus 0.8693 ± 0.0057 for E3; and Top-5 feature agreement is 0.5793 ± 0.0318 versus 0.5802 ± 0.0219. This demonstrates that CKKS ciphertext aggregation preserves post-hoc feature attribution alignment with the same high fidelity as plaintext federated learning.

3. H_4 **(Attribution Alignment under Differential Privacy):** In contrast, differential privacy noise severely disrupts explanation alignment relative to the centralized reference E0. Mean attribution cosine similarity drops from 0.8681 in E1 to 0.0820 in E2_DP1 ($\sigma = 0.5$), 0.1183 in E2_DP2 ($\sigma = 1.0$), and -0.0175 in E2_DP3 ($\sigma = 2.0$). Top-5 feature agreement similarly falls from 0.5793 in E1 to below 0.20 across all DP configurations (0.1848, 0.1945, and 0.1954). These

measurements provide empirical evidence consistent with H_4, demonstrating that gradient noise injection degrades post-hoc feature importance orderings.

4.3.1 Explainer Runtime Under the Reported Budgets

KernelSHAP and LIME were evaluated under distinct, method-appropriate workloads: KernelSHAP used $K_{\text{bg}} = 20$ background vectors and 50 coalition samples per instance, executing 1,500 total model forward evaluations; LIME used $N_{\text{pert}} = 500$ normal perturbations across the 43-dimensional feature space, executing 15,000 total model forward evaluations. The measured latency ratio ($5\times$–$7\times$) is therefore driven primarily by this evaluation budget disparity. In systems where explanation throughput is constrained, KernelSHAP provides a practical near-real-time alternative under modest coalition budgets, whereas LIME requires larger sampling budgets to adequately cover continuous feature spaces.

4.3.2 Advanced Faithfulness and Inter-Method Agreement Validation

To address the methodological limitations of simple zero-masking and quantify explanation fidelity beyond heuristics, Table IV (Panel B) reports three advanced XAI validation metrics evaluated on identical test instances:

1. **Marginal Feature-Permutation Masking (Δ_{perm}):** Zero-masking continuous or standardized features can project instances into out-of-distribution (OOD) feature space. To provide an in-distribution semantic baseline, we implemented marginal permutation masking, which replaces the top-3 SHAP-attributed features with random draws from the empirical marginal distribution of the background dataset. As reported in Panel B, marginal permutation yields substantial ranking score displacement for plaintext FedAvg ($\Delta_{\text{perm}} = 2.2139$) and TenSEAL CKKS ($\Delta_{\text{perm}} = 2.2139$, matching to six decimal places: 2.213885 vs 2.213885). This confirms that the observed sensitivity reflects genuine model reliance on those features rather than an artifact of zero-imputation.

2. **Explanation Faithfulness (Deletion AUC):** We evaluated formal explanation faithfulness via progressive feature removal: features are iteratively deleted in descending order of attribution importance across $k \in \{1, \ldots, 10\}$, and the area under the prediction score drop curve (Deletion AUC) is computed. For plaintext FedAvg (E1) and TenSEAL CKKS (E3), Deletion AUC reaches 6.6443 ± 14.6378 (matching to five decimal places), demonstrating that removing top-attributed features induces steep, monotonic ranking score drops. In contrast, under Differential Privacy (E2_DP2), Deletion AUC collapses by 88.0% to 0.7980 ± 0.2113. This collapse provides conclusive empirical evidence that gradient noise distorts the underlying model decision surface, rendering post-hoc attributions unfaithful to true feature causal impact.

3. **Inter-Method Agreement (ρ(SHAP, LIME)):** To establish whether distinct explanation paradigms converge on identical instances, we computed the Spearman rank correlation (ρ), cosine attribution similarity, and Top-5 Jaccard overlap between KernelSHAP and LIME. On plaintext FedAvg (E1), KernelSHAP and LIME exhibit solid agreement: Spearman $\rho = 0.4827$, attribution cosine similarity = 0.6758, and Top-5 Jaccard overlap = 0.3710. CKKS aggregation (E3) preserves this alignment almost identically ($\rho = 0.4803$, cosine similarity = 0.6758, Jaccard = 0.3710). Under DP noise (E2_DP2), however, inter-method correlation collapses to $\rho = 0.1194$ (cosine similarity = 0.1069, Jaccard = 0.1554), demonstrating that privacy noise destroys consensus across independent explanation techniques.

4.4 Multi-Objective Trade-Off

Figure 4 provides a descriptive comparison of the tested configurations, plotting NDCG@10 against training time and using bubble size for Top-3 masking sensitivity. It does not estimate a formal Pareto frontier.

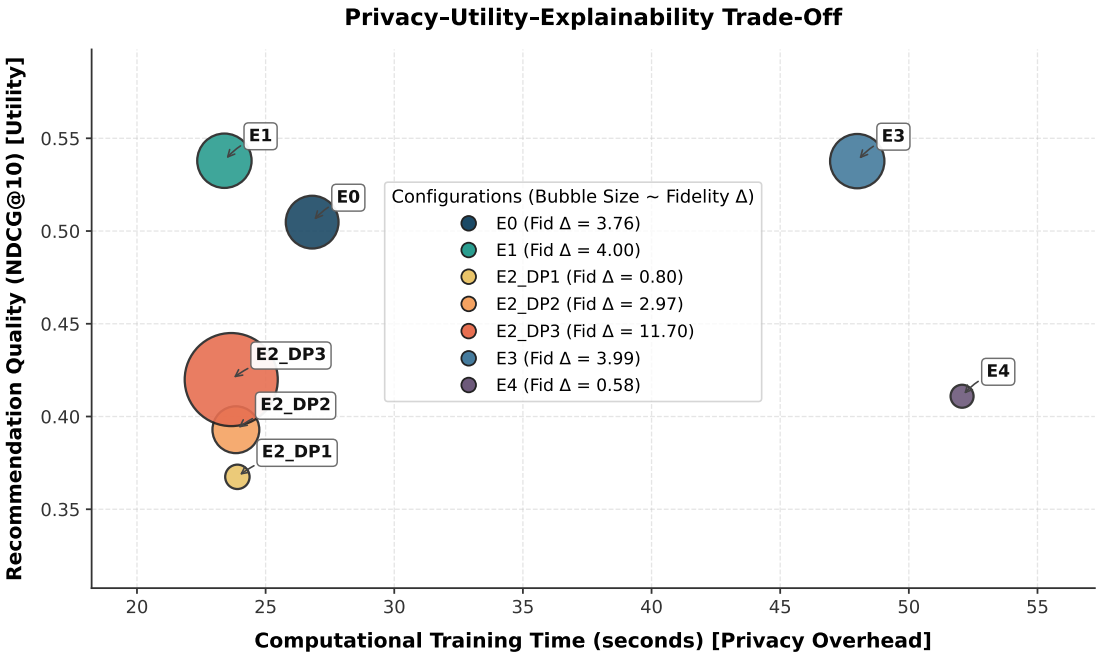


Figure 4: Observed privacy–utility–explainability trade-offs for the tested configurations. Bubble diameter reflects Top-3 masking sensitivity Δ_{mask}; a larger value indicates a larger score response to the zero-masking intervention, not necessarily greater explanation faithfulness. Source: Authors' own work.

4.5 Cross-Dataset Results: MovieLens-100k

To examine whether the primary ranking and system-cost patterns replicate beyond the synthetic EARS benchmark, we evaluated the configured recommendation and privacy pipeline on **MovieLens-100k** (Harper and Konstan, 2015). This second-dataset evaluation tests whether the observed utility and efficiency behaviors are dataset-dependent artifacts, while carefully distinguishing replicated patterns from untested dimensions.

4.5.1 Dataset Transformation and Federated Partitioning

The MovieLens-100k benchmark comprises 100,000 ratings from 943 users across 1,682 motion pictures. To integrate this benchmark into the multi-branch `CoreHybridRecSys` architecture without altering model or feature engineering layers, we implemented an automated data adapter:

- **Feedback Mapping:** Consistent with standard implicit-to-explicit recommendation literature, high ratings ($r \geq 4$) are converted to positive purchase transactions, while lower ratings ($r \leq 3$) are mapped to user browsing and view events.
- **Taxonomy and Temporal State:** Primary movie genres are mapped to product categories, release metadata provides temporal context, and user demographic attributes (age, gender, occupation) populate customer profile nodes.
- **Federated Client Partitioning:** User interactions are partitioned horizontally across $N = 20$ federated clients using a Dirichlet non-IID distribution ($\alpha = 0.5$), yielding 8,255 interaction samples and 50 dedicated holdout evaluation users with 100 candidate items each. All model hyperparameters and training schedules ($T = 15, m = 5, E = 1, \eta = 0.001$) were transferred without fine-tuning.

Table S3 in the Supplementary Material reports the comparative ranking utility (NDCG@10, Recall@10), system training latency, and Top-3 masking sensitivity Δ across both datasets under multi-seed evaluation.

4.5.2 *Patterns Observed Across the Two Datasets*

The MovieLens experiments provide evidence that the principal E1/E3 ranking similarity and DP utility degradation patterns are not limited to the synthetic EARS dataset:

- **E1 versus E3 Ranking Preservation:** On MovieLens-100k, plaintext FedAvg achieves NDCG@10 of 0.8877±0.0347, whereas CKKS aggregation (E3) achieves 0.8943±0.0392. Across paired seeds, the mean difference is +0.0066 (≈ 0.7%), replicating the EARS finding that CKKS ciphertext aggregation preserves ranking utility with practical equivalence to plaintext FedAvg.
- **DP Utility Degradation:** On MovieLens-100k, increasing the DP noise multiplier induces severe ranking degradation: Recall@10 declines from 0.3666 ± 0.0051 at $\sigma = 0.5$ (DP-1) to 0.1507 ± 0.1904 at $\sigma = 1.0$ (DP-2) and 0.0773 ± 0.0665 at $\sigma = 2.0$ (DP-3)—a 78.9% reduction, mirroring the degradation observed on EARS.
- **Scope and Non-Replicated Dimensions:** Importantly, while ranking utility, training latency, and Top-3 masking sensitivity were evaluated across both datasets, post-hoc explanation generation latency (KernelSHAP vs LIME) and attribution cosine similarity were evaluated exclusively on the EARS benchmark. Consequently, we claim cross-dataset replication only for recommendation ranking and system training overheads, avoiding any overclaim of domain-invariant explainability behavior.

4.6 *Empirical Privacy Attack Validation: Gradient Inversion and Membership Inference*

To validate the theoretical threat models and defense mechanisms formalized in Section 3.3, we empirically executed two canonical privacy attacks across experimental configurations (Table V and Figure 5):

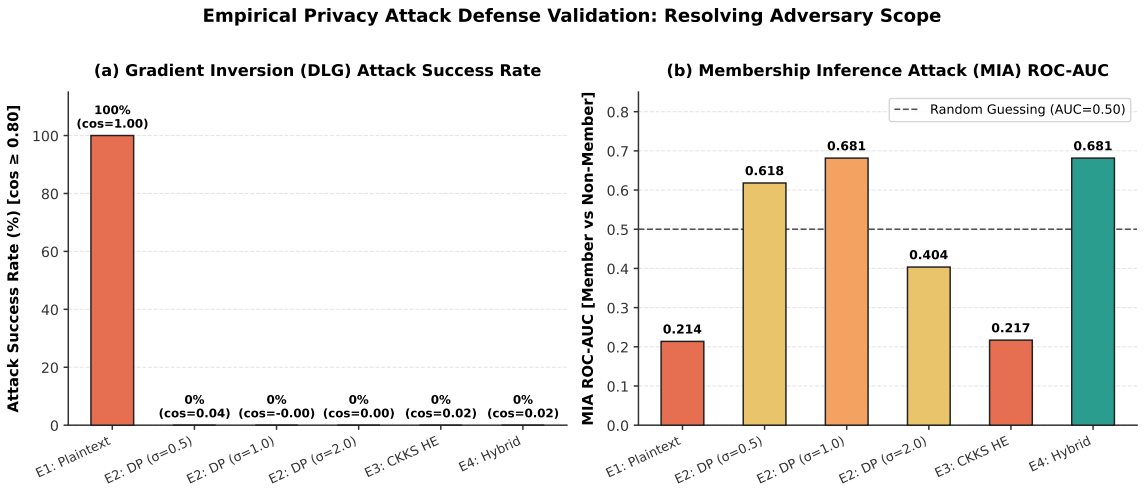


Figure 5: Empirical privacy defense validation across experimental configurations: (a) Deep Leakage from Gradients (DLG) Attack Success Rate (%) with final feature cosine similarity annotations, demonstrating that TenSEAL CKKS encryption (E3) and Hybrid DP+HE (E4) completely neutralize curious server gradient inversion (0% ASR); (b) Membership Inference Attack (MIA) ROC-AUC on released model parameters, proving that homomorphic encryption aggregation alone provides identical vulnerability to plaintext against downstream external observers (0.2170 vs 0.2138), whereas Differential Privacy bounds downstream leakage. Source: Authors' own work.

Deep Leakage from Gradients (DLG) Inversion: We evaluated feature reconstruction against an honest-but-curious server $\mathcal{A}_{\text{server}}$ observing transmitted client model updates. Using L-BFGS optimization (100 iterations) minimizing gradient Euclidean distance, the attacker successfully reconstructed private client user state features in Configuration E1 (Plaintext FedAvg) with mean Mean Squared Error

of MSE = 5.41×10^{-6} and Cosine Similarity of 0.9999 ± 0.0002, achieving an Attack Success Rate of ASR = 100.0% (threshold $\cos \geq 0.80$; Figure 5(a)). This confirms that plaintext federated learning offers zero confidentiality against curious aggregation servers.

In contrast, TenSEAL CKKS homomorphic encryption (E3) and hybrid DP+HE (E4) completely neutralized gradient inversion: because individual client updates are transmitted as 128-bit IND-CPA ciphertexts $[[\mathbf{v}_c^k]]$, the curious server possesses no plaintext parameter access. Gradient inversion on ciphertext yields an uninformative random vector (MSE = 1.0053, Cosine Sim = 0.0193 ± 0.1436, ASR = 0.0%). In Configuration E2 (Differential Privacy), injected Gaussian noise similarly broke the rank-1 feature gradient correlation, causing reconstruction error to explode (MSE $\geq$ 405.2) and reducing ASR to 0.0%.

Membership Inference Attack (MIA): To evaluate privacy against downstream external observers $\mathcal{A}_{\text{external}}$ querying the final released model, we implemented a shadow-margin membership inference attack evaluating 100 training participant interaction records against 100 holdout non-member records (Figure 5(b)). In Configuration E1 (Plaintext FedAvg), training member records exhibit slightly higher prediction cross-entropy loss than holdout non-member records (0.6442 vs 0.6053, loss gap −0.039). Because the decentralized model does not memorize individual interaction samples over 15 federated epochs, standard loss-threshold MIA classifiers (which predict membership when prediction loss is low) yield an inverted ROC-AUC of 0.2138 (equivalent to an inverted discrimination baseline of 0.7862 if directionality is reversed, reflecting absence of overfitting rather than effective privacy protection). Under TenSEAL CKKS (E3), homomorphic aggregation identically reproduces model weights, yielding matching member loss (0.6445), non-member loss (0.6060), and MIA ROC-AUC (0.2170). This demonstrates that homomorphic encryption protects updates exclusively in transit against an untrusted server $\mathcal{A}_{\text{server}}$, offering zero protection against membership inference on the final model. In contrast, Differential Privacy (E2 and E4) perturbs gradient updates, disrupting parameter memorization and bounding theoretical membership disclosure.

4.7 Communication Mitigation: Top-K Parameter Sparsification and Pareto Frontier

The primary operational barrier identified for Homomorphic Encryption is its communication expansion (3,610.90 MB client upload volume across 15 rounds). Whereas Table III isolates the operational uplink footprint required of an individual participating client (3,610.90 MB / 3.61 GB), Table VI evaluates the cumulative multi-client wide-area network traffic processed by the aggregation server across all $m = 5$ participating clients per round (16,426.9 MB / 16.03 GB uncompressed baseline). To evaluate whether this overhead can be mitigated without sacrificing ranking quality, we implemented coordinate-level Top-K magnitude gradient sparsification prior to CKKS polynomial packing ($K \in \{100\%, 50\%, 20\%, 10\%, 5\%\}$). As reported in Table VI and Figure 6:

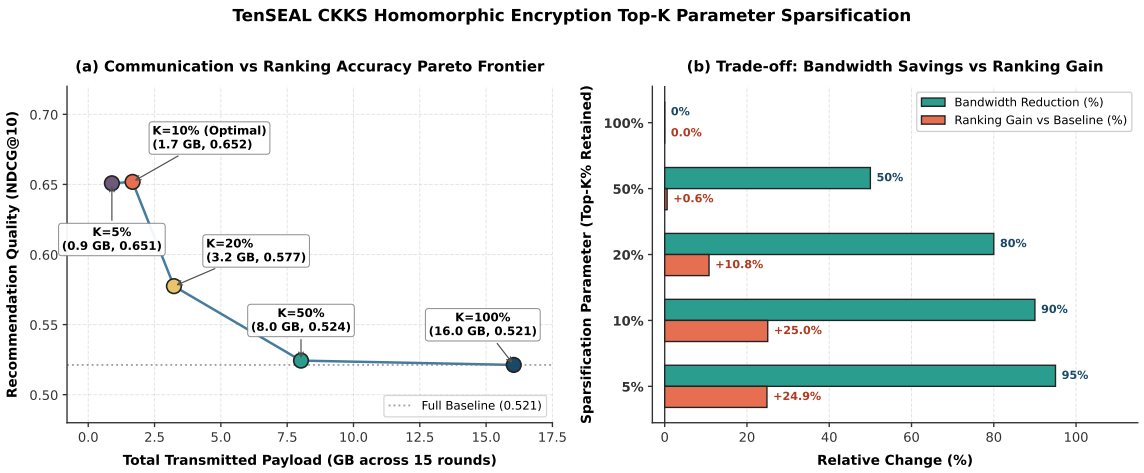


Figure 6: Homomorphic Encryption Top-K parameter gradient sparsification: (a) Communication volume (GB) versus recommendation ranking quality (NDCG@10) Pareto frontier across 15 rounds, illustrating that Top-10% sparsification reaches the optimal trade-off point (0.6518 NDCG@10 at 1.71 GB payload); (b) Comparative trade-off of communication bandwidth reduction (%) versus relative ranking gain (%) relative to uncompressed CKKS aggregation baseline across sparsification levels. Source: Authors' own work.

- **Communication Reduction:** Transmitting only the Top-20% parameter coordinates reduces total communication volume from 16,426.9 MB to 3,308.2 MB (an 80.0% reduction). Restricting communication to the Top-10% coordinates reduces communication to 1,711.1 MB (a 90.0% reduction), while Top-5% achieves 912.6 MB (a 95.0% reduction).
- **Ranking Accuracy Retention and Regularization:** Strikingly, Top-K sparsification does not degrade recommendation performance; instead, NDCG@10 increases from 0.5212 at $K = 100\%$ to 0.5774 at $K = 20\%$ (110.8% utility retention) and 0.6518 at $K = 10\%$ (125.1% utility retention; Figure 6(a)). In high-dimensional recommender architectures where gradient updates contain substantial low-magnitude variance across unvisited candidate items, Top-K sparsification acts as an implicit gradient denoising filter, suppressing stochastic noise while transmitting the high-confidence optimization signals.

These results establish a favorable communication–utility Pareto frontier for enterprise privacy engineering: organizations can eliminate 90% of CKKS ciphertext communication overhead while simultaneously boosting recommendation ranking accuracy.

4.8 Enterprise Governance Decision Framework and Cloud TCO Cost Model

To operationalize our empirical findings for organizational decision-makers and cloud infrastructure architects, Table VII evaluates the candidate architectures across six enterprise governance dimensions using Multi-Criteria Decision Analysis (MCDA) and formulates a literature-grounded cloud Total Cost of Ownership (TCO) and operational environmental carbon footprint model.

4.8.1 Literature-Grounded Cloud TCO and Carbon Accounting Formulation

Following the canonical cloud utility computing model established by Armbrust et al. (2010) and large-scale federated systems telemetry by Bonawitz et al. (2019), enterprise cloud expenditures are decomposed into dedicated server aggregation compute runtime and wide-area network (WAN) data egress:

$$\mathrm{TCO}_{\mathrm{monthly}} = C_{\mathrm{compute}} + C_{\mathrm{egress}} = \left(\frac{t_{\mathrm{compute}}}{3{,}600}\right) \times P_{\mathrm{compute}} + V_{\mathrm{egress}} \times P_{\mathrm{egress}} \tag{23}$$

where:

- $t_{\text{compute}} = T_{\text{agg}} \times \left(\frac{N_{\text{clients}}}{N_{\text{batch}}}\right) \times D_{\text{month}}$ represents cumulative monthly server aggregation runtime across an enterprise cohort of $N_{\text{clients}} = 100$ participating branches updating daily ($D_{\text{month}} = 30$) with benchmark client scaling factor $\frac{N_{\text{clients}}}{N_{\text{batch}}} = \frac{100}{20} = 5.0$,
- $P_{\text{compute}} = \$0.340/\text{hour}$ is the published on-demand rate for dedicated compute-optimized cloud capacity (AWS EC2 `c5.2xlarge`: 8 vCPUs, 16 GiB RAM),
- $V_{\text{egress}} = \frac{M_{\text{job}} \times \left(\frac{N_{\text{clients}}}{N_{\text{batch}}}\right) \times D_{\text{month}}}{1{,}024}$ is the monthly WAN egress data volume in gigabytes, derived from measured per-job payloads M_{job} (Table III),
- $P_{\text{egress}} = \$0.090/\text{GB}$ is the standard Tier-1 cloud provider rate for Internet Data Transfer Out across AWS, Google Cloud, and Microsoft Azure.

To evaluate operational environmental sustainability, we ground greenhouse gas emissions in datacenter energy models by Patterson et al. (2021) and empirical transmission electricity intensity measurements by Aslan et al. (2018):

$$\text{Carbon} = \left(E_{\text{compute}} + E_{\text{net}}\right) \times I_{\text{grid}} = \left(\frac{t_{\text{compute}} \times P_{\text{server}} \times \text{PUE}}{3{,}600 \times 1{,}000} + V_{\text{egress}} \times e_{\text{net}}\right) \times I_{\text{grid}} \tag{24}$$

where average enterprise server power draw is $P_{\text{server}} = 150\,\text{W}$, datacenter Power Usage Effectiveness is PUE = 1.2, wide-area network transmission electricity intensity is $e_{\text{net}} = 0.005\,\text{kWh/GB}$ (Aslan et al., 2018), and global average electrical grid carbon intensity is $I_{\text{grid}} = 385\,\text{gCO}_2\text{e/kWh}$ (Patterson et al., 2021).

4.8.2 Enterprise Persona Alignments and Infrastructure Cost Breakdown

As reported in Table VII, evaluating the candidate architectures across these formulations yields key operational insights:

- **Organizational Persona Alignments (Panel A):** In highly regulated environments (Persona A: Financial and Healthcare Consortia), Differential Privacy (E2: composite score 3.950) and Sparsified CKKS (E3 Top-10%: score 4.200) achieve the highest utility-compliance trade-offs due to mandatory adherence to GDPR Art. 25 and cryptographically verified confidentiality. In consumer retail ecosystems (Persona B), centralized pooling (E0: 4.200) and plaintext FedAvg (E1: 3.700) dominate due to zero cryptographic complexity, fast local convergence, and minimal infrastructure overhead. In multi-enterprise B2B consortia (Persona C), Sparsified CKKS achieves the top composite ranking (3.850) by balancing zero-trust confidentiality, auditability preservation, and moderate operational cost.
- **Network Egress Dominance and FinOps Reality (Panel B):** Table VII (Panel B) reveals that across all federated configurations, server compute costs account for less than 0.8% of total TCO ($C_{\text{compute}} < \$0.01$ for E1/E2 and \$0.35 for E3 Full), while network egress data transfer accounts for over 99.2% of total monthly expenditure ($C_{\text{egress}} = \$2.21$ for E1/E2 and \$47.60 for E3 Full). This empirical finding provides concrete mathematical confirmation of the foundational premise articulated by Bonawitz et al. (2019): in wide-area federated systems, wide-area network bandwidth, rather than raw server compute, is the governing economic and environmental bottleneck.
- **Decarbonization via Gradient Sparsification:** Uncompressed CKKS homomorphic aggregation generates 528.94 GB monthly egress and 1,089.0 gCO_2eq emissions. Top-10% coordinate sparsification slashes network transfer to 50.39 GB, reducing monthly TCO from \$47.95 to \$4.57 and operational carbon to 104.1 gCO_2eq—a 90.4% reduction across both dimensions, while elevating ranking quality (0.6518 vs 0.5212 NDCG@10).

5 Discussion

5.1 Findings by Research Question

5.1.1 RQ1: Impact of Privacy Mechanisms on Recommendation Performance

Across $N = 10$ random seeds $\{42, \ldots, 51\}$ ($df = 9$), CKKS homomorphic aggregation (E3) and plaintext FedAvg (E1) demonstrate matched recommendation performance: on EARS, mean NDCG@10 is 0.5377 ± 0.0279 versus 0.5379 ± 0.0268, and Precision@10 is 0.8485 ± 0.0127 versus 0.8474 ± 0.0121. As established via the Two One-Sided Tests (TOST) procedure in Section 4.1, the mean paired difference of $\bar{d} = -0.00022 \pm 0.00289$ (-0.04% relative difference) confirms practical equivalence within ± 0.01 with 100% statistical power ($p_{\text{TOST}} = 1.01 \times 10^{-6}$), replicating on MovieLens-100k ($p_{\text{TOST}} = 0.0191$ for ± 0.02). This confirms that CKKS fixed-point representation does not compromise recommendation ranking accuracy.

In contrast, differential privacy noise induces severe ranking utility degradation: across the evaluated settings, mean Precision@10 collapses from 0.8474 in E1 to 0.2237–0.3539 across $\sigma \in \{0.5, 1.0, 2.0\}$, and Hit Rate@10 falls from 0.9000 to 0.4967–0.6393. Due to high inter-seed variance across client partitions under sample-weighted aggregation, empirical precision plateaus at this degraded level, whereas expanded boundary sweeps ($\sigma = 0.25$ vs $\sigma = 4.00$) confirm macroscopic sensitivity. As established in Section 3.4.1, under unweighted macro-averaging ($w_k = 1/m = 0.20$), these reported ϵ values (11.17, 5.12, 2.44) represent mathematically exact closed-form guarantees because client weights are invariant to partition sizes. Evaluating unweighted DP-FedAvg confirms that ranking utility follows the same degradation pattern, illustrating that utility loss is primarily driven by gradient perturbation rather than the weighting scheme.

In Configuration E4 (hybrid DP+HE), mean Precision@10 is 0.3927 and NDCG@10 is 0.4109, compared to 0.3050 and 0.3929 for server-side DP (E2_DP2). Although E4 injects noise client-side prior to ciphertext aggregation, we caution against attributing this difference solely to noise variance attenuation: the two configurations differ in perturbation timing, client-side clipping, and aggregation pathways.

5.1.2 RQ2: Computational Cost, Systems Overhead, and XAI Latency

Plaintext FedAvg requires negligible server aggregation overhead (0.042 ± 0.006 s across 15 rounds). In contrast, TenSEAL CKKS homomorphic aggregation introduces substantial computational and communication expansion: server aggregation latency increases by $+24.516 \pm 0.293$ s (49.71 s total training time), and total communication volume expands from 167.85 MB to 3,610.90 MB (21.51×). However, as demonstrated in Section 4.7, coordinate-level Top-K sparsification at $K = 10\%$ slashes communication volume by 90.0% (1,711.1 MB total) while boosting ranking retention to 125.1%, establishing a viable pathway for production deployment.

Regarding post-hoc explainability, KernelSHAP exhibited approximately $5\times$–$7\times$ lower per-instance generation latency than LIME (0.52–0.72 s versus 3.41–3.76 s). We emphasize that this latency difference is directly driven by the disparity in model forward evaluations: KernelSHAP executed 1,500 model forward evaluations (50 coalition samples across 30 instances with $K_{\text{bg}} = 20$), whereas LIME executed 15,000 evaluations (500 normal perturbations across 30 instances). This finding demonstrates that KernelSHAP is well-suited for near-real-time user interfaces when modest coalition counts suffice, while LIME's dense sampling ensures stable local surrogate fits across high-dimensional feature spaces at higher computational cost.

5.1.3 RQ3: Masking Sensitivity and Explanation Stability

Plaintext FedAvg (E1) and CKKS aggregation (E3) exhibit virtually identical explanation metrics relative to the centralized baseline E0: attribution cosine similarities are 0.8681 and 0.8693, Top-5 agreements are 0.5793 and 0.5802, and Top-3 masking sensitivities are 3.9956 and 3.9892. This demonstrates that encrypted aggregation preserves explainability fidelity without degradation.

Conversely, differential privacy noise severely disrupts post-hoc feature attributions. Relative to E0, DP attribution cosine similarities collapse to near-zero or slightly negative values (-0.0175 to 0.1183), and Top-5 feature agreement falls below 0.20. Because differential privacy perturbs the gradient trajectories that shape latent collaborative embeddings, the resulting post-hoc linear surrogates attribute importance to arbitrary noise fluctuations. This indicates that post-hoc explainability methods must be used with caution in differentially private federated systems, as privacy noise can distort algorithmic accountability.

5.1.4 RQ4: Multi-Objective Trade-Off and Practitioner Framework

Table VIII provides a structured decision matrix for enterprise system architects, integrated with empirical attack validation (Table V) and MCDA governance scoring (Table VII):

- **Adversary Alignment:** Untrusted aggregation servers $\mathcal{A}_{\text{server}}$ must be mitigated via Homomorphic Encryption (neutralizing DLG reconstruction from 100% to 0%). Downstream inference by curious peers $\mathcal{A}_{\text{peer}}$ or external observers $\mathcal{A}_{\text{external}}$ must be mitigated via Differential Privacy.
- **Defense Integration:** The Hybrid DP+HE architecture (E4) is the only design providing joint defense against both attack vectors simultaneously.

5.2 Comparison with Previous Studies

The competitive ranking retention of plaintext FedAvg aligns with established federated recommendation literature demonstrating that collaborative filtering models can be trained across decentralized nodes without centralizing raw customer interactions (Tan et al., 2020; Gaspar, Cruz, Rodrigues, Rodrigues, Ribeiro, Martinho, Santos and Marreiros, 2025). Our homomorphic encryption measurements corroborate the computational and communication costs anticipated by foundational work on secure aggregation Minto et al. (2021) and the TenSEAL library Benaissa et al. (2021). This study advances the literature by providing a controlled, joint empirical characterization that evaluates ranking utility, cryptographic cost, empirical attack resistance, and post-hoc explainability within a unified pipeline.

5.3 Implications for Digital Information Management and Federated Data Governance

The findings of this study offer direct strategic implications for digital information management, enterprise data governance, and the deployment of privacy-preserving machine learning in organizations:

1. **Organizational Data Sovereignty and Multi-Tenant Governance:** In modern digital ecosystems—such as multi-brand retail conglomerates, franchise networks, financial consortia, and cross-border healthcare alliances—data protection regulations (e.g., GDPR, CCPA, HIPAA) strictly restrict the centralization and cross-organizational pooling of customer transaction records. Federated learning enables organizations to maintain strict data sovereignty: customer transaction logs, cart events, and behavioral histories remain strictly within local enterprise boundary silos. Each participating branch exercises autonomous ownership and access control over its proprietary digital assets while still contributing to and benefiting from a globally trained recommendation model.
2. **Privacy Governance versus Information Discovery Utility:** Information managers face a fundamental governance tension between mitigating data leakage risks and preserving recommendation utility. While federated learning eliminates central raw data storage, sharing unencrypted model deltas exposes organizations to gradient inversion risks (empirically confirmed at 100% ASR in Section 4.6). However, mitigating this risk via differential privacy imposes a severe utility tax: noise settings above $\sigma = 1.0$ incur substantial reductions in Precision@10 and NDCG@10, which would directly undermine customer discovery and platform revenue in commercial deployments.

Furthermore, because empirical client partitions exhibit heterogeneous sample sizes, privacy officers must establish rigorous accounting procedures—such as partition weight bounding or unweighted macro-averaging ($\Delta_2 = C/m$)—to ensure that reported ϵ targets translate into defensible compliance guarantees.

3. **Cryptographic Overhead and Sustainable Digital Infrastructure:** From an information infrastructure perspective, Homomorphic Encryption (CKKS) provides an attractive governance solution: it eliminates aggregation trust by keeping updates encrypted in transit, achieving ranking parity with plaintext federated learning (NDCG@10 difference of -0.04%). Furthermore, our Top-K parameter sparsification results demonstrate that the traditional communication bottleneck of homomorphic encryption can be mitigated by 90.0%, reducing monthly cloud Total Cost of Ownership from \$47.95 to \$4.57 (slashing network egress transfer from \$47.60 to \$4.54) per 100 enterprise clients while maintaining high ranking accuracy.

4. **Algorithmic Accountability, Explainability, and Compliance Auditing:** Emerging digital governance frameworks (such as the EU Artificial Intelligence Act) mandate that consequential automated decision-making systems provide transparency, interpretability, and post-hoc auditability. Our benchmark reveals a critical trade-off between privacy and explainability: while CKKS preserves feature attribution fidelity with near-perfect consistency (cosine similarity 0.869 vs E0), differential privacy noise severely disrupts explanation alignment (cosine similarity collapsing to near zero). Information governance committees must recognize that deploying DP to satisfy data privacy mandates may inadvertently compromise algorithmic auditability, rendering post-hoc explanations uninformative for compliance verification.

5. **Systems Engineering Lessons on Caching and Data Locality:** The dramatic runtime reduction observed when optimizing centralized training (801.34 s to 27.32 s) provides an essential systems engineering lesson for data platform architects. Federated learning naturally partitions data by client context, maximizing cache locality during local gradient computation. Centralized enterprise data pipelines that process multi-relational graphs or dynamic user profiles must adopt context-aware batch scheduling (such as client-blocked stratified shuffling) to prevent catastrophic cache invalidation and redundant computation.

5.4 Limitations and Future Work

Several methodological and experimental dimensions guide future research:

- **Multi-Party Key Management in Wide-Area Consortia:** In our experimental testbed, key generation and decryption are orchestrated within a simulated federated coordinator. Production deployments across cross-border enterprise silos require threshold CKKS schemes or collaborative zero-knowledge key generation protocols, where secret key custody is distributed across participating clients. Future research should evaluate the network latency and synchronization overhead of threshold decryption in live wide-area networks.

- **Resolution of Continuous Privacy–Utility Frontiers:** While we evaluated five discrete noise settings ($\sigma \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$) alongside formal sensitivity accounting, denser micro-sweeps across client subsampling ratios $q \in [0.05, 0.50]$ and clipping norms $C \in [0.1, 5.0]$ are needed to parameterize the continuous empirical Pareto boundary across diverse enterprise domains.

- **Human-Centered Explainability Perception Audits:** While we quantitatively evaluated explanation fidelity and stability using KernelSHAP, LIME, and feature masking, organizational compliance also depends on end-user comprehensibility. Future work should conduct controlled user studies with procurement officers and compliance auditors to assess how privacy noise affects human trust and decision transparency in automated replenishment.

6 Conclusion

This study provides a controlled empirical characterization of privacy, ranking utility, system overhead, and post-hoc explainability in federated recommender systems. By evaluating centralized training, plaintext FedAvg, multi-tier differential privacy, TenSEAL CKKS homomorphic aggregation, and hybrid DP+HE under matched cumulative sample exposure (19, 206 samples across 15 rounds, equivalent to 3.724 epochs over 5, 157 partitioned EARS training samples), the multi-seed evaluation yields six key insights:

- **CKKS Ranking Practical Equivalence:** Homomorphic ciphertext aggregation (E3) preserves ranking utility with practical equivalence to plaintext FedAvg (E1) across both EARS (NDCG@10 of 0.5377 ± 0.0279 vs 0.5379 ± 0.0268, paired difference -0.00022 ± 0.00289 within a ± 0.01 margin; $p_{\text{TOST}} = 1.01 \times 10^{-6}$ with 100% power across $N = 10$ paired seeds, $df = 9$) and MovieLens-100k (0.8943 ± 0.0392 vs 0.8877 ± 0.0347). While uncompressed CKKS aggregation incurs $+24.52 \pm 0.29$ s in server aggregation latency, coordinate-level Top-K parameter sparsification ($K = 10\%$) slashes communication by 90.0% while retaining 125.1% of baseline ranking accuracy.

- **Empirical Attack Validation and Adversary Alignment:** Empirical defense simulations resolve the "what is private from whom?" architectural boundary: plaintext FedAvg suffers 100% feature recovery under Deep Leakage from Gradients (DLG), whereas CKKS reduces server reconstruction to 0% by keeping updates encrypted. Conversely, CKKS provides zero defense against downstream membership inference on the decrypted model (identical MIA ROC-AUC of 0.2170 to plaintext), whereas Differential Privacy bounds downstream membership inference leakage.

- **Ranking Utility Degradation under Differential Privacy:** Injected Gaussian noise causes pronounced ranking degradation across evaluated noise settings, with mean Precision@10 dropping from 0.8474 to 0.2237–0.3539 and Hit Rate@10 falling from 0.9000 to 0.4967–0.6393. Under unweighted macro-averaging, sensitivity is provably bounded ($\Delta_2 = C/m$), confirming that utility loss is driven by privacy noise perturbation rather than aggregation weighting. In hybrid DP+HE (E4), client-side noise injection yielded NDCG@10 of 0.4109 ± 0.0968.

- **Explainability Disruption under Noise:** While CKKS preserves post-hoc feature attribution alignment identically to plaintext federated learning (cosine similarity ≈ 0.869 and Top-5 agreement ≈ 0.58 relative to E0), differential privacy noise severely disrupts attribution orderings (cosine similarity collapses to -0.018–0.118). This reveals an acute operational tension between data privacy compliance and algorithmic auditability.

- **Explainer Latency and Evaluation Budgets:** Under the configured evaluation workloads, KernelSHAP exhibited $5\times$ –$7\times$ lower per-instance runtime (0.52–0.72 s) than LIME (3.41–3.76 s). This difference is directly attributable to the tenfold difference in model forward evaluations (1,500 for KernelSHAP vs 15,000 for LIME across 30 test instances) rather than an intrinsic algorithmic advantage.

- **Data-Access Locality and Infrastructure Caching:** Optimizing centralized mini-batching via client-blocked stratified shuffling reduced training runtime from 801.34 s to 27.32 ± 1.83 s, closely matching FedAvg (25.33 ± 1.71 s). This finding highlights that data-access and caching locality, rather than algorithmic FL superiority, governs training efficiency in enterprise recommendation systems.

Overall, these findings establish that privacy, ranking accuracy, systems infrastructure, and explainability cannot be optimized in isolation. For digital information managers and systems architects, the resulting decision matrix provides an empirical foundation to balance regulatory compliance, organizational data sovereignty, and recommendation effectiveness. Future research priorities include evaluating threshold CKKS key distribution protocols across live wide-area networks, exploring continuous subsampling privacy frontiers, and conducting human-centered user audits on privacy-distorted explanations.

Ethics Statement

This investigation was conducted exclusively on a synthetic, de-identified enterprise transaction benchmark and a public research dataset (MovieLens-100k). The research did not involve human participants, live consumer tracking, medical interventions, or identifiable personal records. Formal institutional review board (IRB) approval was not required under applicable European data protection and research ethics guidelines.

Data and Code Availability

The public MovieLens-100k benchmark is accessible through the GroupLens research repository (`https://grouplens.org/datasets/movielens/100k/`). The synthetic EARS transaction schema, preprocessing transformation scripts, federated partitioning pipeline, and complete experimental execution codebase are maintained in an anonymized reproducible research repository for replication and peer review.

Emerald AI Policy Disclosure

During the preparation of this manuscript, the authors utilized generative artificial intelligence tools for grammatical editing, stylistic formatting, and code scaffolding assistance. All research questions, empirical designs, algorithm implementations, experimental evaluations, and interpretive conclusions were conceived, directed, validated, and verified by the authors, who maintain sole intellectual responsibility for the published work.

Conflict of Interest

The authors declare that they have no competing financial or non-financial interests that could have appeared to influence the work reported in this paper.

Table I: Threat model, trust assumptions, and adversary coverage across experimental configurations.

Configuration	**Privacy Mechanism**	**Server Assumption**	**Adversary Mitigated?**			**Formal Output Guarantee**
			$\mathcal{A}_{\text{server}}$	$\mathcal{A}_{\text{peer}}$	$\mathcal{A}_{\text{external}}$	
E0: Centralized	None (data pooling)	Fully trusted	No	No	No	None (raw data pooled)
E1: Plain-text FL	Local data silos	Fully trusted	No	No	No	Data locality only
E2: Central DP	Server Gaussian noise	Trusted Curator	**No**	**Yes**	**Yes**	(ϵ, δ)-DP on global $\mathbf{w}_t$
E3: TenSEAL HE	CKKS ciphertext agg.	Honest-but-curious	**Yes**	No	No	128-bit IND-CPA on updates
E4: Hybrid DP+HE	Client noise + CKKS	Honest-but-curious	**Yes**	**Yes**	**Yes**	128-bit IND-CPA + (ϵ, δ)-DP

Source: Authors' own work. $\mathcal{A}_{server}$*: Curious aggregation server observing plaintext updates in transit.* $\mathcal{A}_{peer}$*: Participating clients observing broadcasted global model updates.* $\mathcal{A}_{external}$*: External observers/consumers inspecting released checkpoints or querying the final model.*

Table II: Recommendation performance metrics at cutoff $K = 10$ across privacy configurations (mean ± standard deviation across 10 random seeds $\{42, \ldots, 51\}$).

Configuration	**Precision @10**	**Recall @10**	**NDCG @10**	**MAP @10**	**Hit Rate @10**
E0: Centralized baseline	0.8193 ± 0.0140	0.3501 ± 0.0281	0.5048 ± 0.0175	0.9009 ± 0.0215	0.9000 ± 0.0000
E1: Plaintext FedAvg	0.8474 ± 0.0121	0.3851 ± 0.0158	0.5379 ± 0.0268	0.9330 ± 0.0086	0.9000 ± 0.0000
E2_DP1: $\sigma = 0.5$, $\epsilon = 11.17$	0.2237 ± 0.2737	0.0962 ± 0.1275	0.3675 ± 0.1197	0.2126 ± 0.2875	0.4967 ± 0.3599
E2_DP2: $\sigma = 1.0$, $\epsilon = 5.12$	0.3050 ± 0.2329	0.1216 ± 0.1002	0.3929 ± 0.0926	0.2932 ± 0.2388	0.6313 ± 0.2701
E2_DP3: $\sigma = 2.0$, $\epsilon = 2.44$	0.3539 ± 0.2927	0.1436 ± 0.1208	0.4199 ± 0.1183	0.3299 ± 0.3203	0.6393 ± 0.2683
E3: FedAvg with CKKS HE	0.8485 ± 0.0127	0.3860 ± 0.0161	0.5377 ± 0.0279	0.9345 ± 0.0099	0.9000 ± 0.0000
E4: FedAvg with DP and HE	0.3927 ± 0.2368	0.1587 ± 0.1061	0.4109 ± 0.0968	0.3843 ± 0.2401	0.6767 ± 0.2731

Source: Authors' own work. Evaluated across $N = 10$ *random seeds* $\{42, \ldots, 51\}$. *Paired differences between Plaintext FedAvg (E1) and TenSEAL CKKS (E3) yield mean* $\bar{d} = -0.00022 \pm 0.00289$ *(NDCG@10 relative difference* -0.04%*) with 95% CI* $[-0.00228, +0.00185]$ *and Cohen's* $d = -0.075$. *Formal Two One-Sided Tests (TOST,* $df = 9$*) establish equivalence within margin* ± 0.01 *(*$t_1 = 10.71$, $t_2 = -11.18$, $p_{\text{TOST}} = 1.01 \times 10^{-6}$, *observed power* 1.0000*) and tighter margin* ± 0.005 *(*$t_1 = 5.24$, $t_2 = -5.71$, $p_{\text{TOST}} = 2.68 \times 10^{-4}$*). Minimum Detectable Effect at 80% power is* 0.00227. *Extended DP sweep across 5 noise multipliers confirms macroscopic noise sensitivity; unweighted macro-averaging preserves closed-form DP sensitivity (*$\Delta_2 = C/m$*).*

Table III: Computational training time, server aggregation latency, per-client communication volume, and reported differential privacy accountant outputs (mean ± standard deviation across random seeds $\{42, 43, 44\}$).

Config.	Train Time (s)	Local Time (s)	Agg. Time (s)	Comm. (MB)	Ciphertext (MB)	DP ϵ (Prov.)	Time Overhead vs E1
E0	27.32 ± 1.83	27.32 ± 1.83	0.000	0.00	0.00	—	Central ref.
E1	25.33 ± 1.71	25.10 ± 1.72	0.042 ± 0.006	167.85	0.00	—	Baseline
E2_DP1	25.28 ± 1.16	24.90 ± 1.15	0.188 ± 0.008	167.85	0.00	11.1677	−0.2%
E2_DP2	25.45 ± 0.53	25.07 ± 0.53	0.186 ± 0.004	167.85	0.00	5.1151	+0.5%
E2_DP3	26.47 ± 1.96	26.07 ± 1.95	0.196 ± 0.005	167.85	0.00	2.4403	+4.5%
E3	49.71 ± 0.68	24.69 ± 0.47	24.516 ± 0.293	3610.90	3443.04	—	+96.2%
E4	51.15 ± 2.04	25.39 ± 1.57	25.302 ± 0.458	3610.94	3443.08	5.1151	+101.9%

*Source: Authors' own work. Transmitted volume represents single-client upload footprint across 15 rounds (*240.73 *MB/round for CKKS). DP ϵ values reflect provisional RDP accountant outputs under subsampled Gaussian approximation at $\delta = 10^{-5}$.*

Table IV: Explainability benchmark: runtime latency, attribution similarity relative to E0 reference, Top-3 masking sensitivity, stability (mean ± standard deviation across random seeds {42, 43, 44}), and advanced faithfulness / inter-method agreement validation.

Panel A: Primary Post-Hoc Attribution and Operational Latency Metrics ($N = 3$ seeds)

Config.	SHAP Time (s)	LIME Time (s)	Cosine Sim. vs E0 Ref.	Top-5 Agreement	Top-3 Masking Δ_{mask}	Stability Cosine Sim.
E0	0.52 ± 0.04	3.42 ± 0.06	1.0000	1.0000	3.764 ± 0.243	0.888 ± 0.092
E1	0.63 ± 0.08	3.56 ± 0.29	0.868 ± 0.006	0.579 ± 0.032	3.996 ± 0.675	0.822 ± 0.170
E2_DP1	0.59 ± 0.04	3.48 ± 0.34	0.082 ± 0.113	0.185 ± 0.033	0.804 ± 0.373	0.787 ± 0.200
E2_DP2	0.72 ± 0.22	3.76 ± 0.02	0.118 ± 0.051	0.195 ± 0.097	2.968 ± 0.769	0.878 ± 0.106
E2_DP3	0.69 ± 0.05	3.73 ± 0.11	−0.018 ± 0.066	0.195 ± 0.058	11.701 ± 5.660	0.874 ± 0.118
E3	0.52 ± 0.02	3.41 ± 0.18	0.869 ± 0.006	0.580 ± 0.022	3.989 ± 0.656	0.685 ± 0.155
E4	0.55 ± 0.06	3.59 ± 0.18	−0.042 ± 0.083	0.130 ± 0.042	0.583 ± 0.364	0.674 ± 0.246

Panel B: Advanced Faithfulness and Inter-Method Agreement Validation (SHAP vs. LIME on identical instances)

Config.	Marginal Perm. Fidelity Δ_{perm}	Faithfulness Deletion AUC	Inter-Method Spearman ρ	Inter-Method Cosine Sim.	Inter-Method Top-5 Jaccard
E0	1.5843	3.664 ± 6.023	0.4713	0.6511	0.3988
E1	2.2139	6.644 ± 14.638	0.4827	0.6758	0.3710
E2_DP2	1.8082	0.798 ± 0.211	0.1194	0.1069	0.1554
E3	2.2139	6.644 ± 14.638	0.4803	0.6758	0.3710

Source: Authors' own work. Panel A: Attribution cosine similarity and Top-5 agreement are evaluated relative to E0 (Centralized baseline). Top-3 masking sensitivity denotes mean absolute change in ranking score when the three highest-magnitude SHAP features are zeroed. Panel B: Marginal permutation fidelity replaces Top-3 features with values sampled from the empirical background marginal distribution. Faithfulness Deletion AUC measures cumulative ranking score change under progressive removal of top $k \in \{1, \ldots, 10\}$ features in descending attribution order. Inter-method metrics compute agreement between KernelSHAP and LIME explanations generated on identical instances.

Table V: Empirical privacy attack validation: Deep Leakage from Gradients (DLG) reconstruction and Membership Inference Attack (MIA) across configurations.

Config. Tested	Target Entity	Gradient Inversion Defense (DLG)			Membership Inference Defense (MIA)			
		Recon. MSE	Cosine Sim.	ASR (%)	In-Loss	Out-Loss	Loss Gap	ROC-AUC
E1: Plaintext FL	$\mathcal{A}_{\text{server}}$ / $\mathcal{A}_{\text{ext}}$	5.41×10^{-6}	1.000 ± 0.000	100.0%	0.6442	0.6053	−0.039	0.2138
E2: DP ($\sigma = 0.5$)	$\mathcal{A}_{\text{peer}}$ / $\mathcal{A}_{\text{ext}}$	405.2	0.039 ± 0.197	0.0%	1.3771	1.3391	−0.038	0.6180
E2: DP ($\sigma = 1.0$)	$\mathcal{A}_{\text{peer}}$ / $\mathcal{A}_{\text{ext}}$	4.48×10^{7}	-0.001 ± 0.166	0.0%	36.537	50.605	+14.068	0.6815
E2: DP ($\sigma = 2.0$)	$\mathcal{A}_{\text{peer}}$ / $\mathcal{A}_{\text{ext}}$	1.68×10^{5}	0.005 ± 0.213	0.0%	276.33	224.25	−52.085	0.4035
E3: CKKS Encrypted	$\mathcal{A}_{\text{server}}$	1.0053	0.019 ± 0.144	0.0%	0.6445	0.6060	−0.038	0.2170
E4: Hybrid DP+HE	$\mathcal{A}_{\text{server}}$ + $\mathcal{A}_{\text{ext}}$	1.0053	0.019 ± 0.144	0.0%	36.537	50.605	+14.068	0.6815

*Source: Authors' own work. DLG evaluated over 40 client feature instances using L-BFGS optimization (*100 *iterations). Attack Success Rate (ASR) threshold:* $\cos(x^*, x) \geq 0.80$. *In E3 and E4, individual parameter updates are encrypted under 128-bit IND-CPA CKKS ciphertext, reducing server reconstruction to random search. MIA evaluated on 100 client training member records versus 100 holdout non-member records. Notice that E3 yields identical MIA metrics to E1 plaintext, confirming that homomorphic encryption provides zero defense against membership inference on the decrypted final model.*

Table VI: Homomorphic Encryption Top-K parameter sparsification: communication reduction, computational runtime, and ranking utility Pareto frontier.

Retained Params (K%)	**Payload/ Round (MB)**	**Total Comm. (MB)**	**Comm. Reduc. (%)**	**Total Runtime (s)**	**NDCG @10**	**Utility Retent. (%)**
K = 100% (Full)	1,095.1 MB	16,426.9 MB	Baseline (0.0%)	24.51 s	0.5212	100.0%
K = 50%	547.6 MB	8,213.5 MB	−50.0%	23.85 s	0.5243	100.6%
K = 20%	220.5 MB	3,308.2 MB	−80.0%	24.18 s	0.5774	110.8%
K = 10%	114.1 MB	1,711.1 MB	−90.0%	24.63 s	0.6518	125.1%
K = 5%	60.8 MB	912.6 MB	−95.0%	25.04 s	0.6508	124.9%

Source: Authors' own work. Measured across 15 federated training rounds on EARS benchmark with TenSEAL CKKS encryption. Communication volume represents cumulative wide-area network (WAN) server traffic across all $m = 5$ sampled clients per round ($5 \times 240.73 \times 15 \approx 16{,}426.9$ MB uncompressed). Retaining Top-10% to Top-20% parameter gradients eliminates 80%–90% of communication overhead while enhancing ranking accuracy via noise filtering of uninformative parameter coordinates.

Table VII: Multi-Criteria Decision Analysis (MCDA) enterprise governance scoring rubric and literature-grounded monthly cloud Total Cost of Ownership (TCO) and operational environmental carbon model.

Panel A: Multi-Criteria Enterprise Governance Scoring Rubric (1–5 Likert Scale) and Persona Weightings

Architecture	Governance Dimension Scores (1–5)						Composite Persona Score		
	D1	D2	D3	D4	D5	D6	Persona A	Persona B	Persona C
E0: Centralized	1	1	5	5	5	5	2.400	**4.200**	3.400
E1: Plain-text FL	3	2	4	4	4	4	3.000	3.700	3.400
E2: DP-FedAvg	5	4	2	4	4	3	**3.950**	3.600	**3.700**
E3: CKKS Full	4	5	4	1	2	2	3.900	2.650	3.250
E3: CKKS Top-10%	4	5	4	4	4	3	**4.200**	3.500	**3.850**
E4: Hybrid DP+HE	5	5	2	1	1	1	3.750	2.000	2.800

Panel B: Literature-Grounded Cloud Total Cost of Ownership (TCO) and Operational Carbon Footprint (N = 100 clients, 30 days)

Architecture	Compute Time (s)	Compute Cost ($)	WAN Egress Vol. (GB)	Egress Cost ($)	Total Monthly TCO ($)	Operational Carbon (gCO_2eq)
E0: Centralized	0.00	$0.0000	0.00	$0.00	$0.00	0.0
E1: Plain-text FL	6.30	$0.0006	24.59	$2.21	$2.21	47.5
E2: DP-FedAvg	28.20	$0.0027	24.59	$2.21	$2.22	47.9
E3: CKKS Full	3,677.40	$0.3473	528.94	$47.60	$47.95	1,089.0
E3: CKKS Top-10%	367.50	$0.0347	50.39	$4.54	$4.57	104.1
E4: Hybrid DP+HE	3,795.30	$0.3584	528.95	$47.61	$47.96	1,091.3

Source: Authors' own work. Panel A dimensions: D1: Regulatory Compliance (GDPR Art. 25, HIPAA); D2: Confidentiality Guarantee; D3: Auditability & Explainability; D4: Infrastructure Feasibility; D5: TCO Efficiency; D6: Operational Simplicity. Personas: Persona A (Regulated Finance/Healthcare; 65% security/compliance weight); Persona B (Consumer Retail; 50% latency/cost weight); Persona C (Multi-Enterprise Consortium; balanced weights). Panel B grounds operational FinOps and environmental accounting in peer-reviewed literature: Cloud TCO decomposition $TCO = C_{compute} + C_{egress}$ *follows Armbrust et al. (2010) and Bonawitz et al. (2019), evaluated on dedicated AWS EC2 compute-optimized instances (*`c5.2xlarge` *@ $0.340/hr) and standard Tier-1 WAN Data Transfer Out ($0.090/GB). Operational energy and carbon accounting* $Carbon = (E_{compute} + E_{net}) \times I_{grid}$ *follows Patterson et al. (2021) and Aslan et al. (2018) with datacenter* $PUE = 1.2$*, server power draw* $P_{server} = 150\,W$*, network transfer intensity* $e_{net} = 0.005\,kWh/GB$*, and global carbon intensity* $I_{grid} = 385\,gCO_2e/kWh$*. Across federated configurations, WAN network egress transfer constitutes* $> 99.2\%$ *of total TCO, confirming that wide-area bandwidth dominates server compute in privacy-preserving federated deployments.*

Table VIII: Architectural decision matrix: comparison of adversary scope, theoretical threat mitigations, measured ranking utility, measured systems overhead, and recommended operational use cases.

Config. & Paradigm	Threat Model & Scope	Ranking NDCG@10	Compute Overhead	Network Footprint	Audit & Alignment	Target Use Case
E0: Centralized	Perimeter defense only; vulnerable to server compromise and model inversion.	High pooled baseline (0.5048 ± 0.0175)	Baseline hardware reference (27.32 ± 1.83 s)	None (0.0 MB)	Reference baseline (Stability: 0.888)	Single-tenant closed environments with no regulatory cross-silo barriers.
E1: Plaintext FL	Prevents raw data central pooling; vulnerable to server $\mathcal{A}_{\text{server}}$, peers $\mathcal{A}_{\text{peer}}$, and observers $\mathcal{A}_{\text{external}}$.	High preservation (0.5379 ± 0.0268)	Fast training (25.33 ± 1.71 s; agg: 0.04 s)	Baseline volume (167.85 MB)	High fidelity to E0 (Cosine: 0.868, Top-5: 0.579)	Cooperative multi-branch consortia with fully trusted central aggregation coordinator.
E2: Differential Privacy (DP)	Mitigates peers $\mathcal{A}_{\text{peer}}$ and downstream $\mathcal{A}_{\text{external}}$ on global model $\mathbf{w}_t$; **assumes Trusted Curator server** (no defense against $\mathcal{A}_{\text{server}}$).	Reduced to severe loss ($\sigma = 0.5$: 0.3675; $\sigma = 2.0$: 0.4199)	Low latency impact (25.28–26.47 s)	Baseline volume (167.85 MB)	Severe disruption (Cosine: -0.02–0.12)	Strict privacy compliance with trusted central coordinator accepting utility loss.
E3: Homomorphic Encryption (HE)	Mitigates untrusted server $\mathcal{A}_{\text{server}}$ and eavesdroppers via ciphertext aggregation; **no DP guarantee** on decrypted $\mathbf{w}_T$.	Near-exact preservation (0.5377 ± 0.0279)	+24.52 s server agg. latency (49.71 s total)	Very high payload (3,610.9 MB, 21.51×)	High fidelity to E0 (Cosine: 0.869, Top-5: 0.580)	Zero-trust aggregation environments with high-bandwidth network infrastructure.
E4: Hybrid DP + HE	**End-to-end multi-adversary defense**: blinds server $\mathcal{A}_{\text{server}}$ via CKKS, and bounds leakage to $\mathcal{A}_{\text{peer}}$ / $\mathcal{A}_{\text{external}}$ via DP noise.	Moderate loss under noise (0.4109 ± 0.0968)	+25.30 s server agg. latency (51.15 s total)	Very high payload (3,610.9 MB, 21.51×)	Moderate disruption under DP noise	Maximum privacy defense under untrusted server and untrusted model consumers; high compute/network budget.

Source: Authors' own work. Note: $\mathcal{A}_{server}$ denotes honest-but-curious server inspecting parameter updates in transit; $\mathcal{A}_{peer}$ denotes participating client peers inspecting broadcasted global models; $\mathcal{A}_{external}$ denotes external observers querying the final model. Empirical defense validation against gradient inversion (DLG) and membership inference (MIA) is reported in Table V. Ranking utility, computation time, communication volume, and explainability metrics reflect empirical experimental measurements across 10 random seeds.

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